PHYSICS

1) If a quantity q is given by formula $q = 4t^2 - 2t + \square nt$ then find the value of dq/dt at t = 2.

(1) 11.5
(2) 12.5
(3) 13.5
(4) 14.5

2)

Find the angle between minute hand and hour-hand at 8:50 AM:-

(1) 31°
(2) 60°
(3) 35°
(4) 85°

3) A certain mountain road is inclined 3° with respect to the horizontal. The change in height of the car as a result of its traveling 3 km along the road is nearly :-

(1) 157 m
(2) 181 m
(3) 116 m
(4) 203 m

4)

$y = x^2 - 15x + 56$
for the given equation select the correct options :
(a) Minimum value of y is $-\frac{1}{4}$.
(b) Minimum value of y is $\frac{1}{4}$.
(c) Slope at x = 7.5 is zero.
(d) y = 0 for x = 7 and 8.

(1) b, c, d
(2) a, c, d
(3) b,c
(4) b, d

5) A bird starts from (1, 0, 0) m in the direction $\left(2\hat{i} + 3\hat{j} - 6\hat{k}\right)$ with a speed 21 m/s for 5 sec, then

along the direction $\left(3\hat{i}+4\hat{j}+5\hat{k}\right)$ with a speed $5\sqrt{2}$ m/s for 5 sec. Find the final displacement (in m) of the bird so that it reaches the origin ?

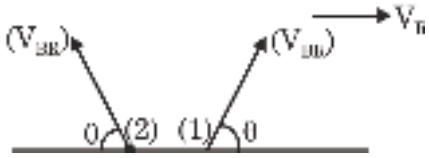
(1) $30\hat{i}+45\hat{j}-45\hat{k}$

(2) $-45\hat{i}-65\hat{j}+65\hat{k}$

(3) $-46\hat{i}-65\hat{j}+65\hat{k}$

(4) $46\hat{i}-65\hat{j}-65\hat{k}$

6) Two boats have same speed w.r.t. river. They head in the direction shown in figure. If T_1 and T_2 are the times taken by boat (1) and (2) respectively to cross the river then :-



(1) $T_1 > T_2$

(2) $T_1 = T_2$

(3) $T_1 < T_2$

(4) Information is not sufficient

7) A particle is situated at the origin of a coordinate system. The following four forces begin to act on the particle simultaneously.

$\vec{F}_1 = \left(2\hat{i}+3\hat{j}-\hat{k}\right)$ N;

$\vec{F}_2 = \left(3\hat{i}-\hat{j}+\hat{k}\right)$ N;

$\vec{F}_3 = \left(2\hat{i}-5\hat{j}+3\hat{k}\right)$ N;

$\vec{F}_4 = \left(5\hat{i}+3\hat{j}-3\hat{k}\right)$ N

The particle will move :-

(1) along x-axis

(2) along y-axis

(3) in x-y plane but not along x or y axis

(4) in y-z plane

8) A particle passes a point moving in north direction with uniform velocity $2\sqrt{3}$ m/s. After some time it turns in the east and moves with uniform velocity 6 m/s. Direction of change in its velocity vector is :

(1) 30° east of north

(2) 60° east of north

(3) 60° south of east

(4) 30° south of east

9) If $\vec{A} = \hat{i}+2\hat{j}+3\hat{k}$ and $\vec{B} = 2\hat{i}-\hat{j}+2\hat{k}$, then find a vector whose magnitude is equal to component of

$\vec{A}$ along $\vec{B}$ and direction is along $\vec{A}$.

(1) $\frac{2\left(2\hat{i}+2\hat{j}+2\hat{k}\right)}{\sqrt{14}}$

(2) $\frac{2\left(2\hat{i}+\hat{j}+3\hat{k}\right)}{\sqrt{14}}$

(3) $\frac{2\left(\hat{i}-2\hat{j}+3\hat{k}\right)}{\sqrt{14}}$

(4) $\frac{2\left(\hat{i}+2\hat{j}+3\hat{k}\right)}{\sqrt{14}}$

10) In a methane (CH_4) molecule each hydrogen atom is at a corner of a regular tetrahedron with the carbon atom at the centre. In coordinates where one of the C–H bonds is in the direction of $\hat{i}+\hat{j}+\hat{k}$, an adjacent C–H bond is in the $\hat{i}-\hat{j}-\hat{k}$ direction. Then angle between these two bonds is :-

(1) $\cos^{-1}\left(-\frac{2}{3}\right)$

(2) $\cos^{-1}\left(\frac{2}{3}\right)$

(3) $\cos^{-1}\left(-\frac{1}{3}\right)$

(4) $\cos^{-1}\left(\frac{1}{3}\right)$

11) A man rows a boat with a speed of 18 km/hr in north–west direction. The shoreline makes an angle of 15° south of west. Obtain the component of the velocity of the boat along the shoreline (in km/hr) :

(1) 9

(2) $18\frac{\sqrt{3}}{2}$

(3) 18 cos (15°)

(4) 18 cos (75°)

12) A particle of mass 10 kg is moving in x-y plane (horizontal) such that its position vector as a function of time is given by
$\vec{r}=-\sqrt{2}\left(\cos t\hat{i}+\cos t\hat{j}\right)$.
The force acting on the particle and path of the particle are respectively :-

(1) $10\sqrt{2}\left(\cos t\hat{i}+\cos t\hat{j}\right)$ & parabolic path

(2) $10\sqrt{2}\left(\cos t\hat{i}+\cos t\hat{j}\right)$ & straight line

(3) $-10\sqrt{2}\left(\cos t\hat{i}+\cos t\hat{j}\right)$ & straight line

(4) $10\sqrt{2}\left(\sin t\hat{i}+\sin t\hat{j}\right)$ & parabolic path

13)

A particle of mass 10 kg is moving in x-y plane (horizontal) such that its position vector as a function of time is given by

$\vec{r}=-\sqrt{2}\left(\cos t\hat{i}+\cos t\hat{j}\right)$.

Distance travelled by the particle from 0 to π sec is

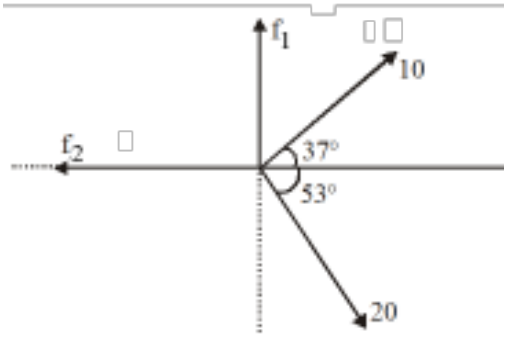
(1) 10 πm

(2) 4m

(3) $2\sqrt{2}\pi m$

(4) $\frac{3}{2}m$

14) Four forces (in N) act on a particle as shown in the figure such that net force is zero. Then select



the **CORRECT** option :-

(a) Magnitude of $\vec{f}_1$ is 10N

(b) Magnitude of $\vec{f}_1$ is 20N

(c) Magnitude of $\vec{f}_2$ is 10N

(d) Magnitude of $\vec{f}_2$ is 20N

(1) a & b

(2) a & c

(3) b & d

(4) a & d

15) Two vectors $\vec{a}$ and $\vec{b}$ have magnitudes of 3 units and 4 units, respectively. Angle between $\vec{a}$ and $\vec{b}$ is θ.

	List-I		**List-II**
(P)	$\vec{a}\cdot\vec{b}=0$	(1)	θ = 0
(Q)	$\hat{a}\cdot\vec{b}=4$ units	(2)	θ = π

(R)	$\vec{a} \cdot \vec{b} = -12$ units	(3)	$\theta = \frac{\pi}{4}$
(S)	$\left\|\vec{a} + \vec{b}\right\| = 5$ units	(4)	$\theta = \frac{\pi}{2}$

(1) P → 4 ; Q → 1 ; R → 2 ; S → 3

(2) P → 4 ; Q → 1 ; R → 2 ; S → 4

(3) P → 3 ; Q → 2 ; R → 1 ; S → 4

(4) P → 2 ; Q → 4 ; R → 1 ; S → 1

16) A point moves such that its displacement as a function of time is given by $x^3 = t^3 + 1$. Its acceleration as a function of time t will be :

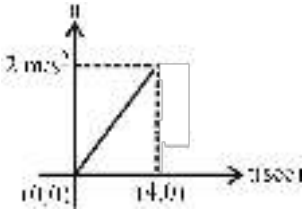
(1) $\frac{2}{x^5}$

(2) $\frac{2t}{x^5}$

(3) $\frac{2t}{x^4}$

(4) $\frac{2t^2}{x^5}$

17) A particle is moving initially with 5 m/s, its a - t curve is as shown in figure. Then its velocity at t = 4 sec is :



(1) 5 m/s

(2) 3 m/s

(3) 2 m/s

(4) 9 m/s

18) A small object is dropped from the top of a building and falls to the ground. As it falls, accelerating due to gravity, it passes window. It has speed v_1 at the top of the window and speed v_2 at the bottom of the window, at what point does it have a speed $\frac{v_1 + v_2}{2}$?
[Neglect the air resistance]

(1) It depends on the height of the window or its distance from the top of the building.

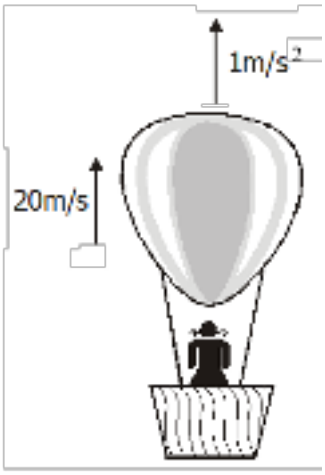
(2) Above the centre point of the window.

(3) Below the centre point of the window.

(4) At the centre point of the window.

19) A balloon is moving along vertical direction with constant upward acceleration of 1 m/s^2. A stone is thrown from the balloon downwards with speed 40 m/s with respect to the balloon. At the time of projection balloon is at height 160 m from the ground and is moving upward with speed 20 m/s. The

time required to fall on the ground by the stone after the projection will be : (Take g = 10 m/s^2)



(1) 8 sec

(2) 4 sec

(3) 10 sec

(4) 2 sec

20) A particle is projected vertically upwards from ground with a velocity 50 m/sec & motion is considered till particle reaches the ground again. Find the correct statement(s) :
(a) It's distance covered in 5th sec & 6th sec will be equal.
(b) For first 4 seconds distance & displacement are same
(c) Displacement in first & last second of the motion are equal.
(d) Distance in first & last second of the motion are equal.

(1) a & c only

(2) a, b & c only

(3) b & d only

(4) a, b & d only

21) A ball is thrown from point O with speed 50 m/s from ground at an angle 53° with horizontal. It lands on same horizontal level at point A. At the instant ball is thrown, a dog starts running with 20 m/s constant velocity along OA, from point O.
Match quantities in List-I with their suitable values in List-II.

	List-I		**List-II (values in appropriate SI unit)**
(P)	Range of ball on ground	(1)	240
(Q)	Horizontal displacement of ball w.r.t. dog after 8 sec.	(2)	80

(R)	Magnitude of velocity of ball w.r.t. dog after 2sec.	(3)	$10\sqrt{5}$
(S)	Magnitude of velocity of ball w.r.t. dog after 5 sec.	(4)	$10\sqrt{2}$

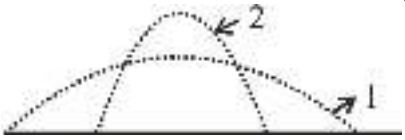
(1) P → 2; Q → 4; R → 1 ; S → 3

(2) P → 1; Q → 2; R → 3; S → 4

(3) P → 3; Q → 2; R → 1 ; S → 4

(4) P → 4; Q → 1; R → 2 ; S → 3

22) Consider the trajectories of two projectiles thrown at different times so that they are not colliding in air. Now choose the correct statements :-



(A) Time of flight of 2 must be greater than that of 1
(B) Horizontal component of velocity of 2 may be greater than that of 1
(C) Initial speed may be equal for the two particles
(D) Initial vertical component of velocity of 1 may be greater than that of 2.

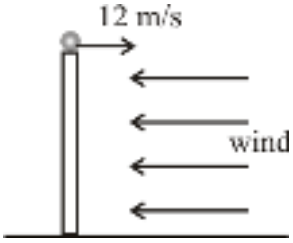
(1) A & D

(2) B & D

(3) A & C

(4) C & D

23) A ball is projected horizontally from the top of a long tower at 12 m/s as shown in figure, wind is blowing as indicated (horizontally), which causes acceleration 6 m/s^2 horizontally. Displacement of the ball when it strikes the tower, (g = 10 m/s^2)



(1) 20 m

(2) 60 m

(3) 80 m

(4) 40 m

24) An aeroplane wants to fly from city-A to city-B, which is 1000 km north of city A. If wind is blowing towards west with speed 300 km/hr and speed of aeroplane is 500 km/hr in still air then,
(a) To reach city-B pilot needs to stear his plane at angle 37° east of north.
(b) To reach city-B pilot needs to stear his plane at angle 53° east of north.
(c) If pilot follows the shortest path, he will reach the city-B in 2.5 hours.
(d) If pilot follows the shortest path, he will reach the city-B in 2 hours.

(1) a & d

(2) b & c

(3) a & c

(4) Only a

25) A hunter tries to hunt a monkey with a small, very poisonous arrow, blown from a pipe with initial speed v_0. The monkey is hanging on a branch of a tree at height H above the ground. The hunter is at a distance L from the bottom of the tree. The monkey sees the arrow leaving the blow pipe and immediately loses the grip on the tree, falling freely down with zero initial velocity. The minimum initial speed v_0 of the arrow for hunter to succeed while monkey is in air :

(1) $\sqrt{\frac{g(H^2+L^2)}{2H}}$

(2) $\sqrt{\frac{gH^2}{\sqrt{H^2+L^2}}}$

(3) $\sqrt{\frac{g(H^2+L^2)}{H}}$

(4) $\sqrt{\frac{2gH^2}{\sqrt{H^2+L^2}}}$

26) Two particles 1 and 2 move with constant velocities $\vec{v}_1$ and $\vec{v}_2$. At the initial moment their position vectors are equal to $\vec{r}_1$ and $\vec{r}_2$. How must these four vectors be inter-related for the particles to collide ?

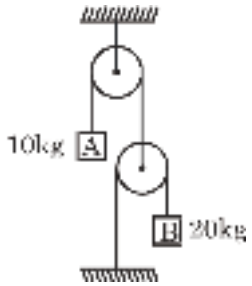
(1) $\frac{\vec{r}_1-\vec{r}_2}{|\vec{r}_1-\vec{r}_2|}=\frac{\vec{v}_1-\vec{v}_2}{|\vec{v}_1-\vec{v}_2|}$

(2) $\frac{\vec{r}_1-\vec{r}_2}{|\vec{r}_1-\vec{r}_2|}=\frac{\vec{v}_2-\vec{v}_1}{|\vec{v}_2-\vec{v}_1|}$

(3) $\frac{\vec{r}_1-\vec{r}_2}{|\vec{v}_2-\vec{v}_1|}=\frac{\vec{v}_2-\vec{v}_1}{|\vec{r}_1-\vec{r}_2|}$

(4) None of these

27) Find the acceleration of block A (Take : g = 10 m/s^2)



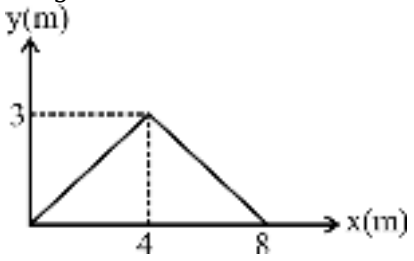
(1) 4 m/s^2

(2) $\frac{10}{3}$ m/s^2

(3) 0 m/s^2

(4) None of these

28) Path of a particle moving in x-y plane is given below. The motion is for 2 sec. The ratio of average velocity and average speed is :-



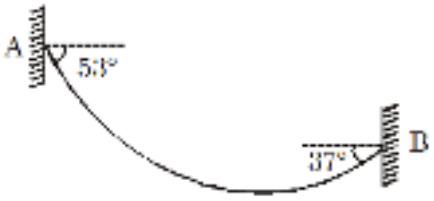
(1) 1

(2) $\frac{4}{5}$

(3) $\frac{3}{4}$

(4) $\frac{1}{2}$

29) A heavy rope is hanging between points A and B. If mass of the rope is 90 kg :

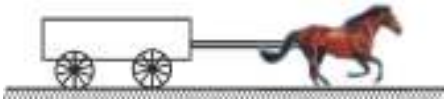


(1) Tension at point A is 900 N

(2) Tension at point B is 540 N

(3) The horizontal components of tension force are different at all points of the rope.

(4) The vertical components of tension force are same at all the points of the rope.

30) A horse is pulling a cart on a rough surface. If we take (horse + cart) as a system, then which of the following statement(s) is/are correct ?  (a) Force by horse on cart is an internal force.
(b) Force by ground on horse is an external force.
(c) Force by cart on horse is an internal force.
(d) Force by ground on cart is an external force.

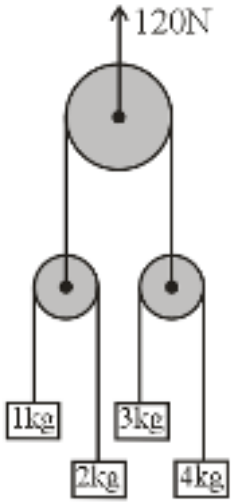
(1) a and b only

(2) c and d only

(3) a, b and c only

(4) a, b, c and d

31) In the given figure all pulleys are smooth and massless and strings are of negligible mass. Match



the information about block in List-I with results in List-II.

List-I		List-II	
(P)	acceleration of 1 kg block.	(1)	Zero
(Q)	acceleration of 2 kg block.	(2)	2.5 m/s^2
(R)	acceleration of 3 kg block.	(3)	5 m/s^2
(S)	acceleration of 4 kg block.	(4)	20 m/s^2

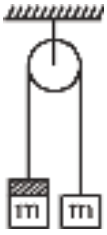
(1) P → 4; Q → 3; R → 1; S → 2

(2) P → 4; Q → 3; R → 2; S → 1

(3) P → 3; Q → 4; R → 1; S → 2

(4) P → 3; Q → 4; R → 2; S → 1

32) Two objects of mass m are connected by a thread that passes over a light, frictionless pulley as shown below. The objects are at rest. If a third object with mass m' is added on top of one of the objects, & objects are released from rest, then the tension in the string becomes $\frac{4}{3}$ times of the initial tension. Find the value of m' :-



(1) $\frac{m}{2}$

(2) m

(3) 2m

(4) 3m

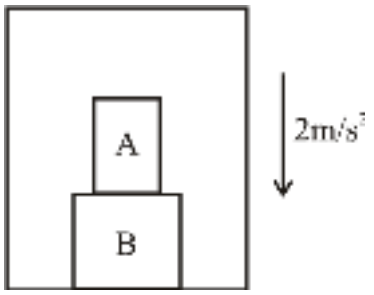
33) A chain consisting of 5 links of mass 0.1 kg each is lifted vertically upwards with a constant acceleration 2.5 m/sec^2 as shown in figure. The force of interaction between the top link and the link



immediately below it will be : (g = 9.8 m/s^2)

(1) 6.15 N
(2) 4.92 N
(3) 3.69 N
(4) 2.46 N

34) The elevator shown in figure descending with an acceleration of 2 m/s^2. The mass of the block 'A' is 5 kg. Force exerted by block 'A' on block B is :- (g = 10 m/s^2)



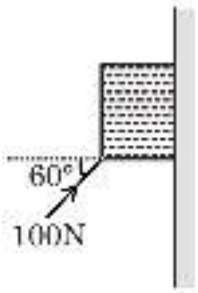
(1) 40 N
(2) 50 N
(3) 10 N
(4) data insufficient

35) A block is placed on horizontal floor. Block is in equilibrium under the action of three forces F_1 = 10N, F_2 = 2N and friction force. If F_2 is removed then net force on block will be :-



(1) 2 N left
(2) 2 N right
(3) 0 N
(4) Cannot be determined

36) A 6.0 kg block initially at rest is pushed against a wall by a 100 N force as shown. The coefficient of kinetic friction is 0.30 while the coefficient of static friction is 0.50. Which of the following is correct about the friction acting on the block :



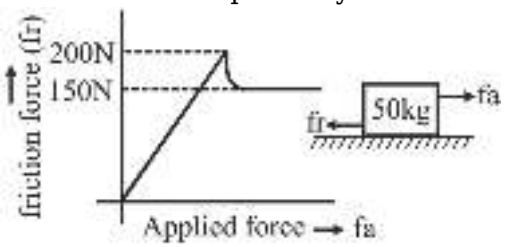
(1) Static friction acts upward on the block.

(2) Kinetic friction acts upward on the block.

(3) Kinetic friction acts downward on the block.

(4) Static friction acts downward on the block.

37) The graph between applied force and frictional force for a body of mass 50 kg placed on a horizontal surface is as shown in figure. Then, what will be the value of coefficient of static and kinetic friction respectively ?



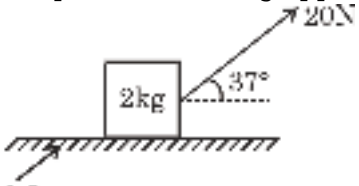
(1) 0.4, 0.3

(2) 0.3, 0.2

(3) 0.4, 0.2

(4) 0.2, 0.3

38) A block lying on a rough surface ($\mu = 0.5$) is being applied by an external force of 20N as shown.



Acceleration of block will be :

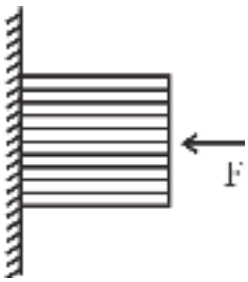
(1) 4 m/s^2

(2) 8 m/s^2

(3) 10 m/s^2

(4) 6 m/s^2

39) How much minimum horizontal force (F) is required to hold a block of mass 2 kg stationary against a wall as shown in the figure :



[Coefficient of friction between the block and the wall is 0.25]

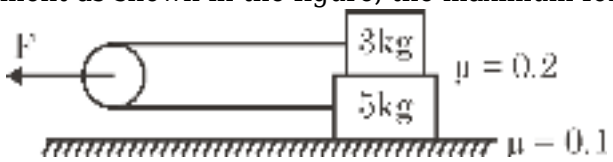
(1) 40 N

(2) 80 N

(3) 8 N

(4) 4 N

40) For the arrangement as shown in the figure, the maximum force F, for no relative motion between blocks, is :



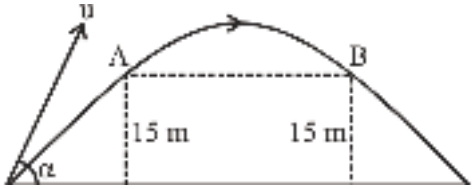
(1) 12 N

(2) 8 N

(3) 20 N

(4) 24 N

41) A golfer standing on level ground hits a ball with a velocity of u = 52 m/s at an angle α above the horizontal. If tan α = 5/12, then the time interval for which the ball is at least 15 m above the ground i.e. between A and B will be : (take g = 10 m/s^2)



(1) 1 sec

(2) 2 sec

(3) 3 sec

(4) 4 sec

42) **Assertion :-** Path of a projected ball is parabolic in uniform gravitational field for oblique projection in absence of air resistance.
Reason :- Gravitational force is always act perpendicular to velocity during the motion of a projectile.

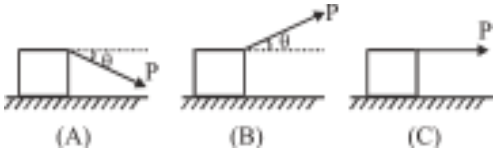
(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Assertion** is false but **Reason** is true.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

43) Figure shows three identical blocks on a rough surface under influence of a force P of same magnitude in all the three cases. Coefficient of friction is same between each block and ground. (Assume that blocks do not overturn about edge and blocks do not slide). Here, f_A, f_B & f_C are frictional forces & N_A, N_B & N_C are normal reactions. Then which option is correct ?



(a) $N_A > N_C > N_B$

(b) $f_A > f_C > f_B$

(c) $f_C > f_A = f_B$

(d) $N_C > N_A = N_B$

(1) Both a and b

(2) Both c and d

(3) Both b and d

(4) Both a and c

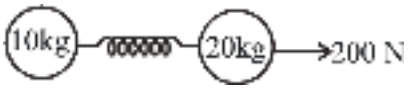
44) A body of mass 1 kg starts moving from origin with initial velocity of $\left(3\hat{i}+4\hat{j}\right)\ ms^{-1}$. If a constant force of $\left(2\hat{i}-3\hat{j}\right)$N starts acting on it, find the coordinate of that point where it reaches in time 3 sec :-

(1) (18, –1.5) m

(2) (9, 1.5) m

(3) (18, 1.5) m

(4) (9, –1.5) m

45) Two masses of 10 kg and 20 kg respectively are tied together by a massless spring. A force of 200 N is applied on a 20 kg mass as shown in figure. At the instant shown, the acceleration of 10kg mass is 12 m/s^2,the acceleration of 20 kg mass is :- 

(1) Zero

(2) 10 m/s^2

(3) 4 m/s^2

(4) 12 m/s^2

CHEMISTRY

1) How many atoms are present in 1 mole of $Ca(OH)_2$?

(1) $30 \times 6.02 \times 10^{23}$ atoms

(2) $5 \times 6.02 \times 10^{23}$ atoms

(3) $6 \times 6.02 \times 10^{23}$ atoms

(4) 6.02×10^{23} atoms

2) Find out the volume of 4.8 g O_3 gas at STP :

(1) 22.4 L

(2) 2.24 L

(3) 0.224 L

(4) 44.8 L

3) The percentage of Se in peroxidase enzyme is 0.5% by mass (atomic mass of Se = 78.4 amu). Then, the minimum molecular mass of enzyme :-

(1) 1.568×10^4 amu

(2) 1.568×10^7 amu

(3) 1.568×10^3 amu

(4) 1.568×10^{6} amu

4) From 280 mg of CO, 10^{21} molecules are removed then number of g molecules of CO remained is :-

(1) $\frac{6}{5} \times 10^{-2}$

(2) $\frac{5}{6} \times 10^{-2}$

(3) 5×10^{21}

(4) 4×10^{21}

5) 15 g of a gaseous hydrocarbon occupy 11.2 L of volume at NTP. It vapour density is :-

(1) 30

(2) 15

(3) 45

(4) 60

6) Calculate the amount of $water_{(g)}$ produced by the combustion of 8g of methane

(1) 36 g

(2) 18 g

(3) 44 g

(4) 22 g

7) What will be the mass of one He atom in gram :-

(1) 6.64×10^{-24} g

(2) 24×10^{23} g

(3) 4.15×10^{-25} g

(4) 6.64×10^{24} g

8) **Assertion:** 1 amu represents a mass equal to 1.66×10^{-27} kg.
Reason: 1 amu represents the actual mass of one atom of carbon-12.

(1) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

(2) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(3) **Assertion** is true but **Reason** is false.

(4) Both **Assertion** & **Reason** are False.

9) How many grams of $AgNO_3$ are needed to prepare 500 mL of a 0.240 M solution ? ($AgNO_3$ = 170 g/mol)

(1) 1.42 g

(2) 20.4 g

(3) 120 g

(4) 354 g

10) 300 mL of a gaseous hydrocarbon when burnt in excess of O_2 gave 2.4 L of CO_2 and 2.7 L of water vapour under same conditions the molecular formula of hydrocarbon is :-

(1) C_4H_8

(2) C_8H_{18}

(3) C_6H_{14}

(4) C_8H_{16}

11) 2 g impure $CaCO_3$ on heating produce 0.84 g of CaO. The percentage purity of $CaCO_3$ sample is (Atomic mass of Ca = 40) $CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

(1) 75%

(2) 84%

(3) 42%

(4) 50%

12) Atomic weight of elements A and B are 30 and 60 respectively. If there are 'y' atoms in the x gm of A. How many atoms would be there in 2x gm of element B :-

(1) y

(2) 2y

(3) 0.5y

(4) 1.5y

13) A given sample of pure compound contains 9.81 g of Zn (atomic mass = 65.39), 1.8×10^{23} atoms of chromium, and 0.60 mol of oxygen atoms. What is the empirical formula ?

(1) $ZnCr_2O_7$

(2) $ZnCr_2O_4$

(3) $ZnCrO_4$

(4) $ZnCrO_6$

14) Match List-I with List-II for the reaction
$C_3H_{8(g)} + 5O_{2(g)} \rightarrow 3CO_{2(g)} + 4H_2O_{(g)}$

	List-I		**List-II**
(a)	8 moles H_2O produced from ____ C_3H_8(g) at NTP	(i)	72g
(b)	88g C_3H_8(g) reacts with ____ O_2	(ii)	3 mole
(c)	44g C_3H_8 produces ____ CO_2	(iii)	44.8L
(d)	1 mole C_3H_8 on reaction with 10 moles O_2 produces _____ H_2O	(iv)	320 g

(1) (a)-(iii), (b)-(iv), (c)-(i), (d)-(ii)

(2) (a)-(i), (b)-(ii), (c)-(iii), (d)-(iv)

(3) (a)-(ii), (b)-(i), (c)-(iii), (d)-(iv)

(4) (a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

15)

What is the molarity of H_2SO_4 solution, that has a density 1.84 g/cc at 35°C and contains solute 98% by mass ?

(1) 4.18 M

(2) 8.14 M

(3) 18.4 M

(4) 18 M

16) With increase in temperature, equilibrium constant of reaction :-

(1) Always increases

(2) Always decreases

(3) May increase or decrease depending upon whether $n_p > n_r$ or $n_p < n_r$

(4) May increase or decrease depending upon whether reaction is endothermic or exothermic

17) On doubling the volume of the vessel which of the following equilibrium will be shifted to right ?

(1) $PCl_5\,(g) \rightleftharpoons PCl_3\,(g) + Cl_2\,(g)$

(2) $H_2\,(g) + Cl_2\,(g) \rightleftharpoons 2HCl\,(g)$

(3) $2CO\,(g) + O_2\,(g) \rightleftharpoons 2CO_2\,(g)$

(4) $N_2\,(g) + 3H_2\,(g) \rightleftharpoons 2NH_3\,(g)$

18) Densities of diamond and graphite are 3.5 and 2.3 g/mL. C (diamond) $\rightleftharpoons$ C (graphite) $\Delta_rH = -1.9$ kJ/mole
favourable conditions for formation of diamond are

(1) high pressure and low temperature

(2) low pressure and high temperature

(3) high pressure and high temperature

(4) low pressure and low temperature

19) For a reaction; $A + B \rightleftharpoons C + D$, the initial concentration of A and B are equal but the equilibrium concentration of C is twice that of equilibrium concentration of A. The K_c is :

(1) 4

(2) 9

(3) 1/4

(4) 1/9

20) For which of the following reactions $K_p < K_c$ at 298 K ?

(1) $N_2O_3(g) \rightleftharpoons NO_2(g) + NO(g)$

(2) $2H_2(g) + O_2(g) \rightleftharpoons 2H_2O(g)$

(3) $H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$

(4) $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$

21) The equilibrium constants for the formation of oxides of nitrogen are given as under (at 25°C):
(a) $N_2 + O_2 \rightleftharpoons 2NO$; $K = 5 \times 10^{-31}$
(b) $N_2 + 2O_2 \rightleftharpoons 2NO_2$; $K = 8 \times 10^{-31}$
(c) $2N_2 + 5O_2 \rightleftharpoons 2N_2O_5$; $K = 1 \times 10^{-34}$
Choose the correct statement :-

(1) NO_2 is the least stable oxide

(2) N_2O_5 is the most stable oxide

(3) The stability order should be : $N_2O_5 > NO_2 > NO$

(4) The stability order should be : $NO_2 > NO > N_2O_5$

22) Consider the reaction; $A_{(g)} + B_{(g)} \rightleftharpoons C_{(g)} + D_{(g)}$; which occurs in one step. The rate constants are 0.25 and 5000 for the forward and reverse reaction respectively. The equilibrium constant is :-

(1) 2.0×10^{-4}

(2) 4.0×10^{2}

(3) 5.0×10^{-5}

(4) 2.5×10^{-6}

23)

The equilibrium constant (K_C) for the reaction
$2HCl(g) \rightleftharpoons H_2(g) + Cl_2(g)$ is 4×10^{-34} at 25°C. What is the equilibrium constant for the reaction
$\frac{1}{2}H_2(g) + \frac{1}{2}Cl_2(g) \rightleftharpoons HCl(g)$

(1) 2×10^{-17}

(2) 2.5×10^{33}

(3) 5×10^{6}

(4) 5×10^{16}

24) **Statement-I :** For the physical equilibrium $H_2O(s) \rightleftharpoons H_2O(\square)$ on increasing temperature and increasing pressure more water will form.
Statement-II : Since forward reaction is endothermic in nature and volume of water is greater than that of volume of ice.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

25) Value of equilibrium constant (K_P) for a reaction
$A(g) \rightleftharpoons 2B(g)$ is 0.0005 atm. What does it indicate ?
(I) Products are more stable than reactants.
(II) Reactants are very less dissociated.
(III) Reactants are more stable than products.
(IV) Value of K_C will be less than K_P at 300 K.

(1) II and III

(2) III and IV

(3) II, III and IV

(4) I, II, III and IV

26) $NH_4HS_{(s)} \rightleftharpoons NH_{3\,(g)} + H_2S_{\,(g)}$
The equilibrium pressure at 25°C is 0.660 atm.
What is K_P for the reaction :-

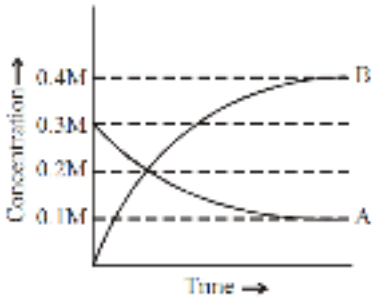
(1) 0.109

(2) 0.218

(3) 1.89

(4) 2.18

27) The figure shows the change in concentration of species A and B as a function of time. The



equilibrium constant K_c for the reaction $A(g) \rightleftharpoons 2B(g)$ is :-

(1) $K_c > 1$

(2) $K_c < 1$

(3) $K_c = 1$

(4) Data insufficient

28)

For the reaction $2AB(g) \rightleftharpoons 2A(g) + B_2(g)$

AB is 50% dissociated at a total equilibrium pressure of P. Then $\frac{P}{K_p}$ will be equal to

(1) 5

(2) $\frac{1}{5}$

(3) 4

(4) $\frac{1}{4}$

29) The value of K_C for the following equilibrium is
$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$.
Given K_p = 167 bar at 1073 K.

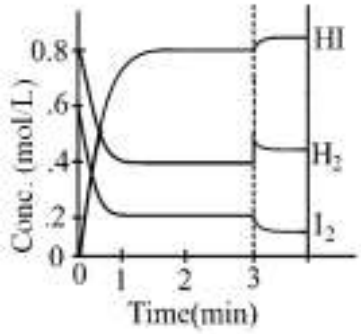
(1) 1.896 mol L^{-1}

(2) 4.38×10^{-4} mol L^{-1}

(3) 6.3×10^{4} mol L^{-1}

(4) 6.626 mol L^{-1}

30) Variation of concentration with time for following reaction in the figure below is:



$H_2(g) + I_2(g)$ – heat ⇌ 2HI(g) At the instant 3 min, what change was imposed into the equilibrium ?

(1) Pressure was increased

(2) Temperature was decreased

(3) Temperature was increased

(4) Hydrogen was added

31) If number of electrons present in x^{+3} is 16, then number of protons in x will be

(1) 16

(2) 13

(3) 18

(4) 19

32) Calculate the frequency of radiation having wavelength 1 Å ?

(1) 3×10^{18} Hz

(2) 3×10^{18} sec^{-1}

(3) 3×10^{18} cycles/sec.

(4) All of the above

33) An element X exist in three isotopic forms X^{40}, X^{41} and X^{42}. If the percentage abundance of X^{40} is 30% and average atomic weight of X is 41.25 then what will be the percentage abundance of X^{41}?

(1) 85 %
(2) 55%
(3) 15%
(4) 35%

34) What will be the diameter of nucleus of an atom which has the mass number 125 :-

(1) 1.66×10^{-11} cm
(2) 1.33×10^{-12} cm
(3) 6.65×10^{-13} cm
(4) 13.3×10^{-12} cm

35) Energy of electron in second orbit of He^{+} is -

(1) –6.80 eV
(2) –13.6 eV
(3) –27.20 eV
(4) –54.4 eV

36) Bohr's model can explain :

(1) the spectrum of hydrogen atom only
(2) the spectrum of atom or ion containing one electron only
(3) the spectrum of hydrogen molecule
(4) the solar spectrum

37) What is the ratio of time periods (T_1/T_2) in third orbit of hydrogen atom to second orbit of He^{+} ion

(1) 8/27
(2) 27/2
(3) 27/32
(4) 27/8

38) A gas absorbs a photon of 355 nm and emits two photons. If one of the emissions is at 680nm, the other is at:

(1) 1035 nm
(2) 325 nm
(3) 743 nm
(4) 548 nm

39) Consider the following statements, the correct statement is :-

(1) Electron density in XY plane in $3d_{x^2-y^2}$ orbital is zero
(2) Electron density in XY plane in $3d_{z^2}$ orbital is zero

(3) 2s orbital has only one spherical node

(4) For $2P_z$ orbital YZ is the nodal plane

40) Incorrect statement is :-

Total energy of electron in third orbit of Li^{+2} ion is

(1) $-\frac{2\pi^2 mk^2e^4}{h^2}$

(2) Potential energy of e^- in second orbit of hydrogen atom is 6.8 eV

(3) Kinetic energy of e^- in third orbit of He^+ is equal to $\frac{8\pi^2 mk^2e^4}{9h^2}$

(4) Kinetic energy of electron in first orbit of hydrogen atom is 13.6 eV

41)

X^+, Y^{2+} and Z^- are isoelectronic to CO_2. Increasing order of protons in X^+, Y^{2+} and Z^- is

(1) $X^+ = Y^{2+} = Z^-$

(2) $X^+ < Y^{2+} < Z^-$

(3) $Z^- < X^+ < Y^{2+}$

(4) $Y^{2+} < X^+ < Z^-$

42) Which of the following statements are correct :-
(a) Two unpaired electrons present in Ni^{+2} ion
(b) One electron present in chromium for which = 0
(c) Penultimate shell of magnesium has 13 electrons
(d) 18 electrons present in the nucleus of Ca^{+2} ion

(1) a, c, d

(2) Only a

(3) a, d

(4) a, b, c, d

43) The uncertainty in position for an electron is $\frac{\lambda}{4\pi}$ where λ is de-Broglie wavelength. The uncertainty in velocity will be (where v is the velocity) :-

(1) $\frac{v}{2}$

(2) v

(3) 3v

(4) $\frac{v}{4}$

44) Which of the following expressions represents the spectrum of Balmer series (if n is the principal quantum number of higher energy level) in Hydrogen atom :-

(1) $\bar{\nu} = \frac{R(n-1)(n+1)}{2n^2}$

(2) $\bar{\nu} = \frac{R(n-2)(n+2)}{4n^2}$

(3) $\bar{\nu} = \frac{R(n-2)(n+2)}{n^2}$

(4) $\bar{\nu} = \frac{R(n-1)(n+1)}{4n^2}$

45) An accelerated electron has a speed of 5×10^6 ms^{-1} with an uncertainity of 0.02%. The uncertainity in finding its location while in motion is $y \times 10^{-9}$ m. The value of y is :-

(1) 38

(2) 48

(3) 68

(4) 58

BIOLOGY



1) Identify given organism and select **true** statement about it ?

(1) *Euglena* ; always heterotrophic

(2) Dinoflagellates ; always autotrophic

(3) *Euglena* ; have two flagella, a short and a long one

(4) Dinoflagellates ; mostly marine and photosynthetic



2) Identify the given diagram and its class :-

(1) *Penicillium* – Ascomycetes

(2) *Agaricus* – Basidiomycetes

(3) *Agaricus* – Ascomycetes

(4) *Penicillium* – Basidiomycetes

3)

Select the incorrect match

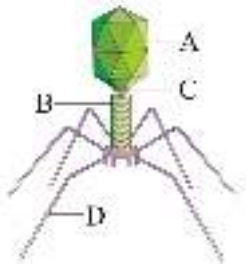
(1) Basidiomycetes - *Puccinia*, *Ustilago*

(2) Ascomycetes - *Penicillium*, *Aspergillus*

(3) Phycomycetes - *Albugo*, *Colletotrichum*

(4) Deuteromycetes - *Trichoderma*, *Alternaria*

4) Given below is the diagram of a bacteriophage. In which one of the options all the four parts A, B, C and D are correct ?



	A	B	C	D
1	Tail fibres	Head	Sheath	Collar
2	Sheath	Collar	Head	Tail fibres
3	Head	Sheath	Collar	Tail fibres
4	Collar	Tail fibres	Head	Sheath

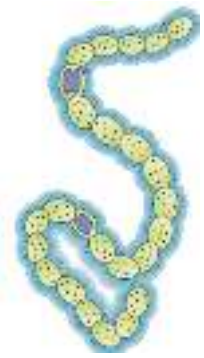
(1) 1

(2) 2

(3) 3

(4) 4

5)



Choose the correct options with respect to given diagram :-

(1) *Nostoc* - Cyanobacteria, can fix N_2.

(2) *Spirillum* - Red algae with motile gamates.

(3) *Anabaena* - Archaebacteria, grow in polluted water.

(4) *Chara* - Brown algae, gives agar.

6) Which of the following structure help bacteria to attach to rocks in streams and to host tissues?

(1) Chromatophores

(2) Flagella
(3) Fimbriae
(4) F-pili

7) Which taxon is not related to wheat :-

(1) Sapindales
(2) Poaceae
(3) Monocotyledonae
(4) *Triticum*

8) The relation of Solanaceae and Convolvulaceae with Polymoniales is similar to the relation occuring in :-

(1) Felidae and Canidae with Carnivora
(2) Primata and Carnivora with Mammalia
(3) Amphibia and Reptilia with Chordata
(4) *Solanum* and *Petunia* with Solanaceae

9) As we go from kingdom to species, the number of common characteristic goes on -

(1) Decreasing
(2) Increasing
(3) Remain same
(4) First increasing then decreasing

10) Which of the following algae represent motile colony ?

(1) *Volvox*
(2) *Spirogyra*
(3) *Porphyra*
(4) *Polysiphonia*

11) On the basis of which criteria algae can be classified into various classes-
(A) Pigment
(B) Morphology of thallus
(C) Stored food
(D) Mode of spore formation and fruiting bodies

(1) A, B, C & D
(2) A & B
(3) C & D
(4) A & C

12) Gemmae are :-

(1) Asexual reproductive structure of pteridophytes

(2) Asexual reproductive structure of liverworts

(3) Sexual reproductive structure of moss

(4) Asexual reproductive spores of algae

13) In which of the following group there is no independent free living existance of gametophyte ?

(1) Thallophyta

(2) Pteridophyta

(3) Bryophyta

(4) Gymnosperm

14) In pteridophytes, spore germinates to produce :-

(1) Sporophyte

(2) Sporangium

(3) Prothallus

(4) Sporocarp

15) Analyse the following characteristics and identify the group of organisms :-
(A) They reproduce sexually by the fusion of non-motile gametes
(B) They show complex post fertilization development.
(C) Majority of organisms are marine with greater concentration found in warmer areas.
(D) They have chlorophyll-a,d and phycoerythrin as major pigments

(1) Phaeophyaceae

(2) Chlorophyaceae

(3) Rhodophyaceae

(4) Dinoflagellates

16) Find out correct statement according to given figure:-



(1) Considered as precurssor to seed habit

(2) It is homosporous plant

(3) Sporangia are formed within Ascocarp

(4) Produces fruit after fertilisation

17) How many plants in the list given below are the members of non vascular plants?
Spirogyra, Volvox, Polysiphonia, Laminaria, Sphagnum, Marchantia, Funaria, Riccia, Moss, Polytrichum, Pteris, Selaginella, Salvinia

(1) Ten

(2) Three

(3) Four

(4) Five

18) Consider the genera given below and answer the question asked next to them :-
Spirogyra, Sargassam, Ulothrix, Laminaria, Ectocarpus, Volvox, Polysiphonia, Funaria, Fucus, Gelidium, Dictyota, Chara
How many above genera belong to green algae and brown algae respectively ?

(1) Six and Five

(2) Four and Five

(3) Three and Four

(4) Five and Six

19) Link between prokaryotes and multicellular eukaryotes is :-

(1) Cyanobacteria

(2) Protista

(3) Fungi

(4) Plants

20) Find out the correct statement-

(1) T.O. Diener discovered a new infectious agent, smaller than virus and named them as prions.

(2) Viroids lack the protein coat and made of only free RNA

(3) The RNA of viroids is of high moecular weight

(4) Lichens are very good pollution indicator as they grow only in polluted areas.

21) Study the following statement and give the answer:-
(A) Bacteria are sole members of the kingdom monera
(B) Bacteria are most abundant micro-organisms
(C) Bacteria are generally larger and multiply more rapidly than eukaryotic cells
(D) In bacteria, mesosome help in respiration
(E) Bacteria show most extensive metabolic diversity

(1) A, B & E are correct

(2) A, C & E are correct

(3) A, B, D & E are correct

(4) A, B, C, D and E are correct

22) Match the column :

	Column-I		**Column-II**
(a)	Family	(i)	*tuberosum*
(b)	Kingdom	(ii)	Polymoniales

(c)	Order	(iii)	*Solanum*
(d)	Specific epithet	(iv)	Plantae
(e)	Genus	(v)	Solanaceae

(1) a-iv, b-v, c-ii, d-i, e-iii

(2) a-v, b-iv, c-ii, d-i, e-iii

(3) a-v, b-iv, c-i, d-iii, e-ii

(4) a-i, b-ii, c-iii, d-iv, e-v

23) Match the column :-

Column-I		**Column-II**	
A	Binomial nomenclature	(i)	Aristotle
B	Father of biology & zoology	(ii)	Linnaeus
C	Darwin of 20th century	(iii)	Theophrastus
D	Father of botany	(iv)	Ernst Mayr

(1) A-ii, B-i, C-iv, D-iii

(2) A-ii, B-iv, C-i, D-iii

(3) A-iii, B-iv, C-i, D-ii

(4) A-i, B-iii, C-ii, D-iv

24) Select the odd one out with respect to functions of endomembrane system -

(1) Involved in protein synthesis and secretion

(2) Synthesis of lipids and steroids

(3) Active r-RNA synthesis

(4) Intracellular and intercellular digestion

25) Select the incorrect statement :

(1) Microbodies contain various enzyme and are present in both plant and animal cells

(2) Telocentric chromosome have a terminal centromere

(3) Nucleolus is a structure present in nucleoplasm and it is a site for ribosomal RNA synthesis

(4) Every chromosome essentially has a secondary constriction

26) (A) Typically mitochondria are sausage shaped or cylindrical shaped
(B) The ribosomes of mitochondria are bigger than cytoplasmic ribosomes.

(1) A is correct, B is incorrect

(2) A is incorrect, B is correct

(3) Both A and B are correct

(4) Both A and B are incorrect

27) Why the golgi body (GB) remains in close association with the endoplasmic reticulum (ER) ?

(1) GB and ER are doule membrane bounded structure.

(2) GB and ER are membraneless structure.

(3) Proteins synthesised by ribosomes on the ER are modified in the cisternae of the GB.

(4) Origin of both cell organelles is common.

28) How many of the following statements are true?
(i) Chromatin contains DNA, RNA, histones and non-histones.
(ii) A single human chromosome has approximately two meter long DNA.
(iii) Every chromosome essentially has a primary constriction.
(iv) Nucleolus is a site for active ribosomal RNA synthesis.
(v) The outer membrane of nucleus does not bear ribosomes on it.

(1) Two

(2) Three

(3) Four

(4) Five

29) Centrioles differ from cilia or flagella :-
(a) In number of microtubules in each peripheral fibre.
(b) In number of radial spokes.
(c) In arrangement of microtubules.
(d) In containing proteinaceous central hub.
(e) In being surrounded by double membrane.
Options :

(1) a, c and d

(2) a, b, c and d

(3) a, b and c

(4) b, c, d and e

30) Read the following statements & select the correct statements.
(a) Pachytene stage is characterised by the appearance of recombination nodules.
(b) In telophase decondensation of chromosomes occurs.
(c) Nucleolus is not a membrane bound structure.
(d) Microbodies are present only in animal cells.
(e) Mitochondria are the site of photo phosphorylation.
Choose the correct options :

(1) a, b & c

(2) b, c & d

(3) a, c & d

(4) a, b & d

31) Match the column-I, II and III :

	Column-I		**Column-II**		**Column-III**

x	Robert Brown	a	Final shape to cell theory	i	Eukaryotic cell
y	Rudolf Virchow	b	Ribosome	ii	Both Eukaryotic & Prokaryotic cells
z	George Palade	c	Discovery of Nucleus	iii	All cells arise from pre existing cells

Options :

(1) x-a-i, y-b-ii, z-c-iii

(2) x-c-ii, y-b-i, z-a-iii

(3) x-c-i, y-a-iii, z-b-ii

(4) x-b-iii, y-c-ii, z-a-i

32) **Assertion (A) :** Chloroplasts and mitochondria are semi-autonomous cell organelles.
Reason (R) : Both of these organelles have their own genetic material and protein synthesis machinery.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is not correct.

(3) (A) is incorrect but (R) is correct.

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A).

33)

How many cell organelles in the list given below are without membrane :-

Golgibody, Vacuole, Nucleolus, Lysosome, Mitochondria, Peroxisomes, Glyoxysomes, Centriole, Chloroplast, Ribosome.

(1) One

(2) Two

(3) Five

(4) Three

34) Which of the following is correctly matched ?

(1) SER – Synthesis of protein

(2) RER – Synthesis of steroidal hormone

(3) Golgibody – Formation of nuclear membrane

(4) Lysosome – Digestion of nucleic acid

35) **Assertion :-** The cytoplasm is the main arena of cellular activities in cell.
Reason :- The various chemical reactions occur in the cytoplasm to keep the cell in the living state.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

36) Identify the following :



(1) A-Telocentric, B-Metacentric, C-Acrocentric, D-Sub-metacentric

(2) A-Telocentric, B-Acrocentric, C-Sub-metacentric, D-Metacentric

(3) A-Telocentric, B-Metacentric, C-Sub-metacentric, D-Acrocentric

(4) A-Metacentric, B-Telocentric, C-Acrocentric, D-Sub-metacentric

37)

Which of the following structure is similar in bacteria and plant cell?

(1) Cell wall

(2) Ribosomes at cisternae of ER

(3) Cell membrane

(4) Nuclear membrane

38) What do (i) and (ii) represent in the given flow chart ?

Parent cell $\xrightarrow{M-I}$ Two Daughter $\xrightarrow{M-II}$ Four Daughter
(2n) cells (i) cells (ii)

(1) (i) = 2n, (ii) = n

(2) (i) = n, (ii) = n

(3) (i) = n, (ii) = 2n

(4) (i) = 2n, (ii) = 4n

39) Identify the meiotic stage in which the homologous chromosomes separate while the sister chromatids remain associated at their centromeres :

(1) Anaphase I

(2) Anaphase II

(3) Metaphase I

(4) Metaphase II

40) Identify the **correct** sequence of events during Prophase I of meiosis :

(a) Synapsis of homologous chromosomes
(b) Chromosomes become gradually visible under microscope
(c) Crossing over between non-sister chromatids of homologous chromosomes
(d) Terminalisation of chiasmata
(e) Dissolution of synaptonemal complex.
Choose the **correct answer** from the options given below :

(1) (a), (b), (c), (d), (e)

(2) (b), (c), (d), (e), (a)

(3) (b), (a), (c), (e), (d)

(4) (a), (c), (d), (e), (b)

41) Given below are two statements : one is labelled as **Assertion (A)** and the other is labelled as **Reason (R).**
Assertion (A) : The growth of multicellular organism is due to mitosis.
Reason (R) : Mitosis is also called as equational division and it offers genetic stability.
In the light of the above statements, choose the **correct** answer from the options given below :

(1) **(A)** is not correct but **(R)** is correct

(2) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

(3) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(4) **(A)** is correct but **(R)** is not correct

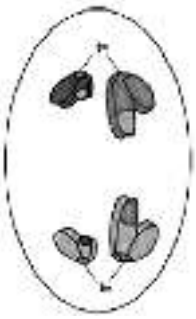
42) Choose the correct sequence of stages of Prophase I :

(1) Leptotene → Zygotene → Pachytene → Diplotene → Diakinesis

(2) Leptotene → Pachytene → Zygotene → Diplotene → Diakinesis

(3) Leptotene → Zygotene → Pachytene → Diakinesis → Diplotene

(4) Leptotene → Pachytene → Zygotene → Diakinesis → Diplotene

43)  Identify the above stage with its respective event :-

(1)	Prophase II	splitting of the centromere
(2)	Anaphase I	Homologous chromosomes separate

(3)	Anaphase I	Splitting of the centromere
(4)	Metaphase I	Bivalent chromosomes align on the equatorial plate

(1) 1

(2) 2

(3) 3

(4) 4

44) If a cell has 10 Pg DNA in G_1-stage of interphase, what will be the amount of DNA in each daughter cell after meiosis-I :-

(1) 10 pg

(2) 20 pg

(3) 40 pg

(4) 5 pg

45) What will be the number of chromatids in prophase I of a cell having 20 bivalents?

(1) 20

(2) 80

(3) 40

(4) 10

46) Match the columns :-

	Column-I		**Column-II**
A.	Simple epithelium	i.	Some of the columnar or cuboidal cells get specialized for secretion.
B.	Glandular epithelium	ii.	lining for body cavities, ducts and tubes.
C.	Adhering Junction	iii.	Perform cementing to keep neighboring cells together
D.	Squamous epithelium	iv.	Made of single thin layer of flattened cells with irregular boundaries.

(1) A–iii, B–ii, C–iv, D–i

(2) A–iv, B–iii, C–i, D–ii

(3) A–ii, B–i, C–iii, D–iv

(4) A–iv, B–i, C–iii, D–ii

47) **Statement I :-** Connective tissues is the most abundant and widely distributed in the body of complex animals.

Statement II :- Connective tissues range from a soft connective tissue to specialized types which include cartilage, bone and blood.
Choose the correct option from the given below.

(1) Both Statement I and Statement II are incorrect.

(2) Statement I is correct but Statement II is incorrect.

(3) Statement I is incorrect but Statement II is correct

(4) Both Statement I and Statement II are correct.

48) How many of the following statements is/are correct about epithelium ?
(a) It is called epithelial tissue.
(b) It usually doesn't have free surface
(c) It's main function is to hold various body surfaces together.
(d) Cells of epithelium have a large amount of intercellular matrix between them.

(1) Two

(2) Three

(3) One

(4) Four

49) Given below are two statements one is labelled as Assertion (A) and the other is labelled as Reason (R)
Assertion (A) :- The intercellular material of cartilage is solid and pliable and resists compression. Cells of this tissue (chondrocytes) are enclosed in small cavities within the matrix secreted by them
Reason (R) :- Most of the cartilages in vertebrate embryos are replaced by bones in adults. Cartilage is present in the tips of nose, outer ear joints, between adjacent bones of the vertebral column, limbs and hands
In the light of above statement, choose the correct answer from the options given below :-

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true and R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

50) Here, a classification of Indian frog is given. Find the best terms to fill the blanks.

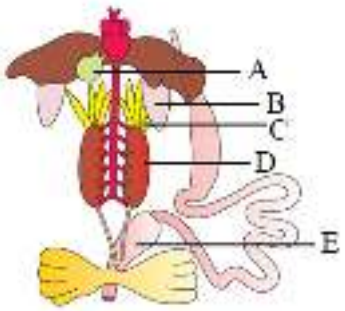
Kingdom	Animalia
Phylum	Chordata
____(i)____	Amphibia
order	____(ii)____
____(iii)____	*Rana*
Species	____(iv)____

(1) (i)-Class, (ii)-Anura, (iii)-Genus, (iv)-*tigrina*

(2) (i)-Family, (ii)-Tetrapoda, (iii)-Order, (iv)-*Bufo*

(3) (i)-Class, (ii)-Anura, (iii)-Family, (iv)-*Bufo*

(4) (i)-Family, (ii)-Tetrapoda, (iii)-Genus, (iv)-*tigrina*



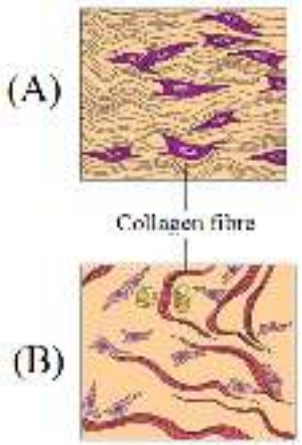
51) Find out the correct match for labelling shown in the figure :

(1) A-Gall bladder - synthesis of bile

(2) B-Liver - secretion of bile.

(3) D - Kidneys-Ducts from it opens in urinary bladder.

(4) D-Kidneys - possess uriniferous tubules



52) Select the correct statement about the given diagrams

(1) In **B**-collagen fibres are present in rows between many parallel bundles of fibres

(2) In **A**-fibroblast and many fibres are oriented differently

(3) Example of **A** is Tendon

(4) **B** is an example of specialised C.T.

53)

Column-I		**Column-II**	
(a)	Endocrine glands	(i)	Earwax
(b)	Exocrine glands	(ii)	Dry surface of skin
(c)	Compound epithelium	(iii)	Hormones
		(iv)	Saliva

		(v)	Mucus
		(vi)	Moist surface of buccal cavity

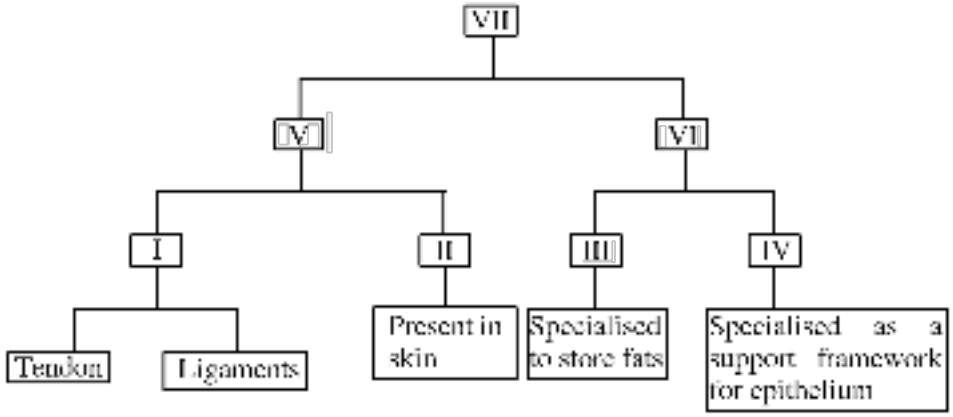
(1) a-(iii); b-(i), (iv), (v); c-(ii), (vi)

(2) a-(ii), (iv); b-(iii), (vi); c-(i), (v)

(3) a-(v); b-(ii), (iii); c-(i), (iv), (vi)

(4) a-(ii), (vi) b-(iii); c-(i), (iv), (v)

54) Recognise the following flow diagram and find out the correct option.



	(1)	(2)	(3)	(4)
I.	Dense regular tissue	Dense irregular tissue	Dense regular tissue	Dense irregular tissue
II.	Dense irregular tissue	Dense regular tissue	Dense irregular tissue	Dense regular tissue
III.	Areolar tissue	Areolar tissue	Adipose tissue	Adipose tissue
IV.	Adipose tissue	Adipose tissue	Areolar tissue	Areolar tissue
V.	Loose connective	Loose connective tissue	Dense connective tissue	Dense connective tissue
VI.	Dense connective	Denser connective tissue	Loose connective tissue	Loose connective tissue
VII.	Connective tissue	Connective tissue	Connective tissue	Connective tissue

(1) 1

(2) 2

(3) 3

(4) 4

55) The midbrain of frog is characterised by :

(1) Two pairs of optic lobe

(2) One optic lobe

(3) Three optic lobes

(4) A pair of optic lobes

56)

Read the following four statements about sense organs of "Rana tigrina" :
(a) Frog has simple eyes possessing only one unit.
(b) External ear is present in frog.
(c) External ear is absent in frog and only tympanum can be seen externally.
(d) Eyes and internal ear are well organised structures and the rest are cellular aggregation.
How many of the above statements are correct?

(1) 4

(2) 1

(3) 2

(4) 3

57) The site of final digestion in frog is :-

(1) Stomach

(2) Rectum

(3) Oesophagus

(4) Intestine

58)

Which of the following statements is incorrect regarding digestion in frogs?

(1) Stomach contains HCl and proteolytic enzymes.

(2) Digestion of food takes place by the action of HCl and gastric juice secreted from the wall of the stomach.

(3) Digestion is completed in stomach.

(4) Partially digested food is called chyme.

59)

Given below is a list of few structures:
Diaphragm, Tail, Neck, Cloaca, Teeth on Upper jaw, Nictitating membrane
How many of the above structures are found in “Rana tigrina”?

(1) 4

(2) 5

(3) 6

(4) 3

60) Which of the following statements is not correct for Indian bull frog ?

(1) Bidder's canal in male frog communicates with the urinogenital duct.

(2) In female frog oviduct and ureter not open independently in cloaca.
(3) There is no functional connection between ovary and kidney of frog.
(4) Bidder's canal present in the kidney of male frog.

61) In male frogs, the copulatory pad is present on the :

(1) Second digit of the fore limbs
(2) First digit of the hind limbs
(3) First digit of the fore limbs
(4) Second digit of the hind limbs

62) The function which is mainly performed by compound epithelium is -

(1) Secretion
(2) Absorption
(3) Diffusion
(4) Protection

63) Match column I with column II

Column-I		**Column-II**	
A	Squamous epithelium	i	Multilayered
B	Compound epithelium	ii	Single layer of flattened cells
C	Areolar tissue	iii	Diffusion boundaries
		iv	Dry surface of skin
		v	Beneath the skin
		vi	Support for epithelium

(1) A-i, iii ; B-ii, iv ; C-v, vi
(2) A-iii, iv ; B-ii, v ; C-i, vi
(3) A-v, vi ; B-i, ii ; C-iii, iv
(4) A-ii, iii ; B-i, iv ; C-v, vi

64) In the given table fill tissue and related organ in blanks :-

Tissue	**Organ**
Squamous epithelium	Walls of blood vessels
__i__	Tubular parts of nephron
__ii__	Lining of stomach
Ciliated epithelium	__iii__

(1) i-Columnar epithelium, ii-Cuboidal epithelium, iii-Bronchioles

(2) i-Squamous epithelium, ii-Cuboidal epithelium, iii-Fallopian tube

(3) i-Ciliated epithelium, ii-Cuboidal epithelium, iii-Bronchioles

(4) i-Cuboidal epithelium, ii-Columnar epithelium, iii-Fallopian tube

65) In the urinary bladder, the epithelium shows the ability to stretch when urinary bladder is full and contract when it is empty. The cells appear cuboidal in a relaxed state but flattened when stretched. Such unique structural adaptation is characteristic of :

(1) Stratified squamous epithelium

(2) Transitional epithelium

(3) Simple cuboidal epithelium

(4) Ciliated columnar epithelium

66) Match the following :

(I)	*Pila*	(A)	Devil fish
(II)	*Pinctada*	(B)	Apple snail
(III)	*Loligo*	(C)	Pearl oyster
(IV)	*Octopus*	(D)	Squid

(1) I–D, II–B, III–C, IV–A

(2) I–A, II–B, III–C, IV–D

(3) I–B, II–C, III–D, IV–A

(4) I–C, II–D, III–B, IV–A

67) Identify wrong statement with regard to phylum platyhelminthes :

(1) They are mostly ectoparasite

(2) They are bilaterally symmetrical

(3) Their body is dorso-ventrally flattened

(4) They are triploblastic animals.

68) Which phylum is triploblastic acoelomate ?

(1) Ctenophora

(2) Platyhelminthes

(3) Aschelminthes

(4) Coelenterata

69) Which features are common to the animals belonging to class-amphibia and class-reptilia.

(1) The presence of scales with internal fertilisation and usually four chambered heart

(2) The presence of cloaca, oviparous and external fertilisation

(3) Skin is moist

(4) The presence of tympanum, poikilotherms and usually three-chambered heart.

70) In which one of the following, the genus name, its two characters and its phylum are not correctly matched whereas the remaining three are correct ?

	Genus Name	**Two characters**	**Phylum**
(1)	*Pila*	(i) Body segmented (ii) Mouth with radula	Mollusca
(2)	*Asterias*	(i) Spiny skinned (ii) Water vascular system	Echinodermata
(3)	*Sycon*	(i) Pore bearing (ii) Canal system	Porifera
(4)	*Periplaneta*	(i) Jointed appendages (ii) Chitinous exoskeleton	Arthropoda

(1) 1

(2) 2

(3) 3

(4) 4

71) In tissue level of organisation the :-

(1) Cells are arranged as loose cell aggregate

(2) Tissue are grouped to form organs

(3) Cells performing the same function are arranged into tissues

(4) Tissues are grouped to form systems

72) Which is not a correct match :

(1) *Chelone* – Turtle

(2) *Calotes* – Garden lizard

(3) *Bangarus* – Viper

(4) *Testudo* – Tortoise

73)

Operculum and air bladder are absent in :-

(1) *Torpedo*

(2) *Labeo*

(3) *Hippocampus*

(4) *Clarias*

74) Which of the following statement is correctly associated with cyclostomata ?

(1) Sucking & circular mouth with jaws
(2) Paired fins present
(3) Cranium & vertebral column are cartilaginous
(4) 4 - 6 pairs gills present for respiration

75) Find out the correct set of viviparous animals :-

(1) *Platypus, Pteropus, Camelus*
(2) *Canis, Felis, Elephas*
(3) *Ornithorhynchus, Canis, Felis*
(4) *Delphinus, Platypus*

76) Which coelenterate has exoskeleton composed of calcium carbonate :-

(1) *Aurelia*
(2) *Meandrina*
(3) *Corals*
(4) Both (2) and (3)

77) Three chambered heart, cold blooded, moist skin and indirect development are characteristics of :-

(1) *Labeo*
(2) *Pavo*
(3) *Salamandra*
(4) *Corvus*

78) Which of the following cells line the spongocoel in poriferans ?

(1) Choanocytes
(2) Interstitial cells
(3) Gastrodermal cells
(4) Nematocysts

79) In which of the following animals blood circulates in a closed circuit through a series of blood vessels of varying diameter ?

(1) *Pila*
(2) *Pheretima*
(3) *Periplaneta*
(4) *Balanoglossus*

80) Which among the following is the correct combination of aquatic mammals ?

(1) Dolphins, Penguins
(2) Penguins, Duck billed Platypus

(3) Dolphins, Whales

(4) Penguins, Ostrich

81) Which of the following organisms neither have notochord nor vertebral column in adult stage ?

(1) *Myxine*

(2) *Petromyzon*

(3) *Herdmania*

(4) *Scoliodon*

82) **Assertion :** Arthropoda is the largest phylum of kingdom animalia.
Reason : Over two-thirds of all named animal species on earth are arthropodos.
Choose the most appropriate answer from the options given below :-

(1) Both assertion and reason are true and the reason is a correct explanation of the assertion.

(2) Both assertion and reason are true but the reason is not a correct explanation of the assertion.

(3) Assertion is true but the reason is false.

(4) Assertion is false but the reason is true.

83) Which group of animals belong to the same phylum ?

(1) Earthworm, Roundworm, Tapeworm

(2) Prawn, Scorpion, Locusta

(3) Sponge, Sea anemone, Starfish

(4) Penguin, Dolphin, Star fish

84) Which of the following set of animal all are not homeotherms ?

(1) *Calotes, Chelone, Clarias*

(2) *Pteropus, Psittacula, Struthio*

(3) *Corvus, Delphinus, Elephas*

(4) *Equus, Macaca, Balanoptera*

85) Which of the following pairs of animals are similar to each other pertaining to feature stated against them ?

(1) *Pteropus* and *Ornithorhyncus* – viviparity

(2) Garden lizard and crocodile – 4 chamber heart

(3) *Ascaris* and *Ancylostoma* – Metameric segmentation

(4) Sea horse and Flying fish – Poikilothermal

86) In which of the following groups, all animals are monoecious ?

(1) Tapeworm, *Neries, Pheretima*

(2) Leech, Tapeworm, Earthworm

(3) *Ascaris*, Tapeworm, Earthworm

(4) Star fish, *Pheretima, Sepia*

87) Given below are the list of animals, read it carefully :-
Rana tigrina, Labeo, Naja, Columba, Pristis, Bufo, Hyla, Scoliodon, Exocoetus
Out of these how many animals are poikilothermal and having three chambered heart ?

(1) Six

(2) Seven

(3) Eight

(4) Four

88) *Nereis* differ from leech in :-
(a) Paropodia
(b) Excretion by nephridia
(c) Metamerically segmented
(d) Dioecious
(e) Bilateral symmetry

(1) a, b, d

(2) a, d only

(3) b, d, e

(4) b, c, e

89) Read the given statements and select the correct option :-
Statement-I : All triploblastic animals are eucoelomates.
Statement-II : Diploblastic animals have a false coelom.

(1) Both statement I and II are incorrect.

(2) Both statement I and II are correct.

(3) Statement-I is not correct but statement-II is correct.

(4) Statement-I is correct but statement-II is not correct.

90) Which of the following has internal fertilization ?

(1) Frog

(2) *Ctenoplana*

(3) *Balanoglossus*

(4) *Scypha*

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	4	3	1	2	3	2	1	4	4	3	1	2	2	4	2	2	4	2	2	4
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	2	3	3	3	1	2	2	2	2	4	1	2	2	1	4	3	1	4	2	4
Q.	41	42	43	44	45															
A.	2	2	4	1	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	2	1	2	2	2	1	3	2	2	1	1	2	4	3	4	1	3	1	2
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	4	3	4	2	3	1	1	1	1	4	4	4	3	2	2	2	2	3	3	2
Q.	86	87	88	89	90															
A.	3	2	2	2	4															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	2	3	3	1	3	1	1	2	1	4	2	4	3	3	1	1	2	2	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	2	1	3	4	1	3	2	1	1	3	4	2	4	1	2	3	2	1	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	1	2	1	2	3	4	3	2	1	4	3	1	3	4	4	4	3	4	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	3	4	4	4	2	3	1	2	4	1	3	3	1	3	2	4	3	1	2	3
Q.	171	172	173	174	175	176	177	178	179	180										
A.	3	1	2	1	4	2	4	2	1	4										

SOLUTIONS

PHYSICS

1)

$q = 4t^2 - 2t + \square nt$

$i = \frac{dq}{dt} = 8t - 2 + \frac{1}{t}$

at t = 2 sec

$i = (8 \times 2) - 2 + \left(\frac{1}{2}\right)$

i = 14.5 amp

2)

08 : 50 AM = 8 hour + 50 minutes

= 8 × 60 minutes + 50 minutes

= 480 minutes + 50 minutes

= 530 minutes

Hour hand :

 Angle made by hour hand in 1 minute = $\left(\frac{1}{2}\right)^0$

 Angle made by hour hand in 530 minutes

$= \frac{1}{2} \times 530 = 265^0$

Minute hand :

 Angle made by min hand in 1 minute = 6°

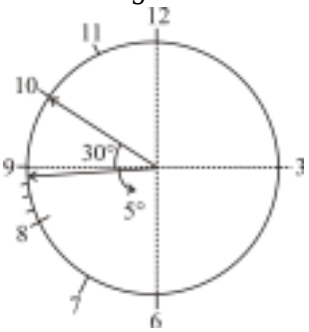
 Angle made by min hand in 530 minutes

= 6° × 530 = 3180°

= 8 × 360° + 300°

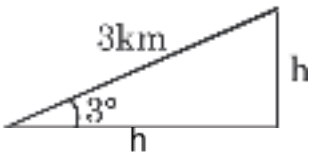
= 8 rounds + 300°

Hence angle between minute hand & hour hand will be ⇒ θ = 300° – 265° = 35°



Angle is ⇒ 30° + 5° = 35°

3)



$\sin\theta = \frac{h}{3000}$

for small angle $\sin\theta \Rightarrow \theta$

$\frac{3 \times \pi}{180} = \frac{h}{3000}$

h = 157 m

4)

Roots :

$x^2 - 15x + 56 = 0$

$(x - 7)(x - 8) = 0 \Rightarrow x = 7,8$ (Option d is correct)

Minimum Value (Vertex):

$x = -\frac{b}{2a} = -\frac{(-15)}{2(1)} = 7.5$

$y_{min} = (7.5)^2 - 15(7.5) + 56 = 56.25 - 112.5 + 56$

$= -0.25 = -\frac{1}{4}$ (Option a is correct)

Slope:

$\frac{dy}{dx} = 2x - 15$

Slope at

x = 7.5 : 2(7.5) – 15 = 15 – 15 = 0. (Option c is correct)

so ans. a,c,d

5)

$\vec{r}_i = \hat{i}$

$\vec{d}_1 = 21 \times \left[\frac{2\hat{i} + 3\hat{j} - 6\hat{k}}{7}\right] \times 5 = 30\hat{i} + 45\hat{j} - 90\hat{k}$

$\vec{d}_2 = 5\sqrt{2} \times \frac{\left[3\hat{i} + 4\hat{j} + 5\hat{k}\right]}{5\sqrt{2}} \times 5 = 15\hat{i} + 20\hat{j} + 25\hat{k}$

$\vec{d}_{Net} = 45\hat{i} + 65\hat{j} - 65\hat{k}$

$\vec{r}_f = 46\hat{i} + 65\hat{j} - 65\hat{k}$

$\vec{d}_f = -\vec{r}_f = -\left(46\hat{i} + 65\hat{j} - 65\hat{k}\right) = -46\hat{i} - 65\hat{j} + 65\hat{k}$

6) Component of velocity ⊥ to flow (v sinθ) is same for both boat.

Hence $T_1 = T_2$

7) $\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 = 12\hat{i}$

⇒ Particle moves along x-axis.



8)

$$\tan\theta = \frac{2\sqrt{3}}{6} = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}$$

$\Rightarrow \theta = 30°$ south of east.

9)

$$\text{Required vector} = \frac{\vec{A}.\vec{B}}{|\vec{B}|} \times (\text{Unit vector along A})$$

$A \cdot B = (1\times2) + (2\times-1) + (3\times2) = 6$

$|B| = \sqrt{(4+1+4)} = 3$

Component of A along B = 6/3 = 2

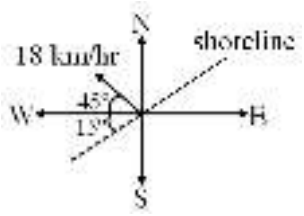
$$\text{Unit vector along A} = \frac{(\hat{i} + 2\hat{j} + 3\hat{k})}{\sqrt{14}}$$

$$\text{Required vector} = \frac{2(\hat{i} + 2\hat{j} + 3\hat{k})}{\sqrt{14}}$$

10) $\cos\theta = \dfrac{\vec{A}.\vec{B}}{AB}$

$$\cos\theta = \frac{(\hat{i}+\hat{j}+\hat{k}).(\hat{i}-\hat{j}-\hat{k})}{\sqrt{1+1+1}\sqrt{1+1+1}} = \frac{1-1-1}{3} = \frac{-1}{3}$$

$$\theta = \cos^{-1}\left(\frac{-1}{3}\right)$$



11)

∵ angle of velocity with shoreline is 60°.

∵ component of v is $u\cos\theta$

$\Rightarrow (18)(\cos(60^\circ))$

$$\Rightarrow 18 \times \frac{1}{2} = 9 \text{ km/hr}$$

12) $\vec{r} = -\sqrt{2}\left(\cos t\hat{i} + \cos t\hat{j}\right)$

$$\vec{v} = \frac{d\vec{r}}{dt} = \sqrt{2}\left(\sin t\hat{i} + \sin t\hat{j}\right)$$

$$\vec{a} = \sqrt{2}\left(\cos t\hat{i} + \cos t\hat{j}\right)$$

$$\vec{F} = m\vec{a} = 10\sqrt{2}\left(\cos t\,\hat{i} + \cos t\,\hat{j}\right)$$

Angle between velocity & acceleration is

$$\cos\theta = \frac{2(2\sin t\cos t)}{4\sin t\cos t} = 1$$

$\cos\theta = 1$

$\theta = 0°$

So path is straight line

13)

Position vector is given by $\vec{r} = -\sqrt{2}\left(\cos t\,\hat{i} + \cos t\,\hat{j}\right)$

 Velocity $\vec{v} = \frac{d\vec{r}}{dt} = -\sqrt{2}\left(-\sin t\,\hat{i} - \sin t\,\hat{j}\right)$

$$\vec{v} = \sqrt{2} \times \sin t\left(\hat{i} + \hat{j}\right)$$

 Speed of particle = 2 sin t

So, distance travelled in 0 to π second.

$$d = \int_0^{\pi} 2\sin t\,dt = 2[-\cos t]_0^{\pi}$$

$$= -2[\cos\pi - \cos 0]$$

Distance = 4m

14)

$f_1 = -10\sin 37° + 20\sin 53°$

$f_2 = 10\cos 37° + 20\cos 53°$

15) (P) $\vec{a}\cdot\vec{b} = ab\cos\theta = 0$

$$\therefore\theta = \frac{\pi}{2}$$

(Q) $\hat{a}\cdot\vec{b} = b\cos\theta = 4$

$\Rightarrow 4\cos\theta = 4$

$\therefore \theta = 0$

(R) $\vec{a}\cdot\vec{b} = -12 = 3\times 4\cos\theta$

$\therefore \theta = \pi$

(S) $\left|\vec{a}+\vec{b}\right| = 5 = \sqrt{3^2+4^2}$

$$\therefore\theta = \frac{\pi}{2}$$

16) $x^3 = t^3 + 1 \Rightarrow 3x^2\frac{dx}{dt} = 3t^2$

$x^2v = t^2 \Rightarrow 2x\frac{dx}{dt}v + x^2a = 2t$

$$\Rightarrow 2x\frac{t^4}{x^4} + x^2a = 2t \Rightarrow x^2a = 2t - \frac{2t^4}{x^3}$$

$$\Rightarrow a = \frac{2t[x^3 - t^3]}{x^5} \Rightarrow a = \frac{2t}{x^5}$$

17) $\int a\,dt = \Delta v$

$\int a\,dt \rightarrow$ area under a – t curve

$\frac{1}{2}(4)(2) = v_f - v_i$

$v_f - v_i = 4 \rightarrow v_f = 4 + 5 = 9$ m/s

18) Body accelerates downwards so the speed increases linearly with time. But the distance fallen increases as t^2. So the average speed occurs at half of time taken to pass the window, which is before it has covered half of the height of the window.

19) $v_{bs} = 40$ m/s downwards

$v_b = 20$ m/s upwards

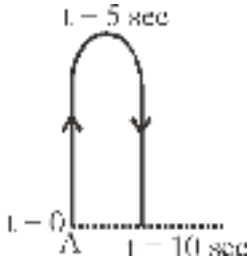
$v_b = -40 + 20 = -20$ (downwards)

Initial velocity of stone is 20 m/s downwards

$\Rightarrow -160 = -20t - 5t^2$

or $t = 4$ or -8 (neglect the negative value)

$\Rightarrow t = 4$ seconds.

20) 

(a) distance travelled in 5th second and 6th second are 5 m both.

(b) in first 4 second direction of velocity does not change so distance = displacement (c & d) displacement is a vector quantity. Distance in first and last second are equal but displacement not.

Hence answer (4)

21) (P) $T = \frac{2u\sin\theta}{g}$

T = 8 sec

$R = \frac{u^2\sin 2\theta}{g}$

R = 240 m

(Q) w.r.t. dog horizontal velocity of ball

= 10 m/s

So disp = 10 × 8 = 80 m

(R) at t = 2s

$\vec{v}_{b/G} = 30\hat{i} + 20\hat{j}$

$\vec{v}_{dog/G} = 20\hat{i}$

$|\vec{v}_{b/d}| = \left|10\hat{i} + 20\hat{j}\right| = 10\sqrt{5}$ m/s

(S) at t = 5s

$\vec{v}_{b/G} = 30\hat{i} - 10\hat{j}$
$\vec{v}_{d/G} = 20\hat{i}$
$|\vec{v}_{b/d}| = \left|10\hat{i} - 10\hat{j}\right| = 10\sqrt{2}$ m/s

22)

$(H_{max})_2 > (H_{max})_1$
$(u_y)_2 > (u_y)_1$
$T_2 > T_1$
$u_x = \frac{R}{T}$
$(u_x)_1 > (u_x)_2$

$u = \sqrt{u_x^2 + u_y^2}$ may be equal for both

23) Ball strikes towers ⇒ $S_x = 0$

$0 = 12 \times t + \frac{1}{2}(-6)t^2$ ⇒ t = 4 sec

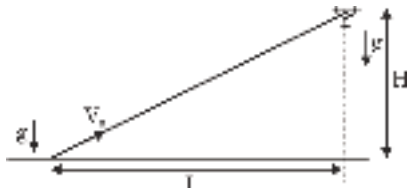
So displacement in y-direction in 4 sec

$S_y = \frac{1}{2} \times 10 \times 16 = 80m$

24) $\vec{v}_{P/g} = \vec{v}_{P/w} + \vec{v}_w$



$\sin\theta = \frac{3}{5} \Rightarrow \theta = 37°$ $v_{P/g}$ = 400 km/hr

25) 

$t = \frac{\sqrt{H^2 + L^2}}{V_0}$

$V_0 \to min \quad t \to max$

$H = \frac{1}{2}gt_{max}^2$

$V_{0min} = \frac{\sqrt{H^2 + L^2}}{t_{max}} = \sqrt{\frac{g\left(H^2 + L^2\right)}{2H}}$

26) Position vector of particle-1 at time 't'

$= \vec{r}_1 + \vec{v}_1 t$

Position vector of particle-2 at time

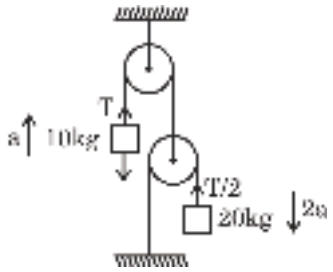
't' $\rightarrow \vec{r}_2 + \vec{v}_2 t$

For collision, their position vectors should become equal at some time 't'

$\vec{r}_1 + \vec{v}_1 t = \vec{r}_2 + \vec{v}_2 t$

$(\vec{r}_1 - \vec{r}_2) = (\vec{v}_2 - \vec{v}_1)\, t$

So unit vectors of $(\vec{r}_1 - \vec{r}_2)$ and $(\vec{v}_2 - \vec{v}_1)$ should be same.



27)

T – 10g = 10a ....(i)

$20g - \frac{T}{2} = 20(2a)$ ....(ii)

28)

Initial position (0,0) final position (8,0)

Displacement: $\Delta\vec{r} = \vec{r}_f - \vec{r}_i = \left(8\hat{i} + 0\hat{j}\right) - \left(0\hat{i} + 0\hat{j}\right) = 8\hat{i}\text{m}$

$|\Delta\vec{r}| = 8\text{m}$

Total Distance : The particle moves from (0,0) to (4,3)

distance covered $= \sqrt{(4-0)^2 + (3-0)^2} = \sqrt{16+9} = 5\text{m}$

The particle moves from (4,3) to (8,0)

distance covered $= \sqrt{(8-4)^2 + (0-3)^2} = \sqrt{16+9} = 5\text{m}$

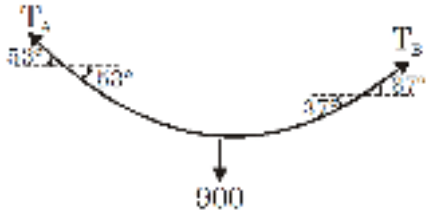
Total distance travelled = 5 + 5 = 10 m

Average velocity $= \frac{\text{Total displacement}}{\text{Total time}} = \frac{8}{2} = 4$ m/s

Average speed $= \frac{\text{Total distance}}{\text{Total time}} = \frac{10}{2} = 5\text{m/s}$

Ratio of Average velocity to average speed

$\frac{\text{Average velocity}}{\text{Average speed}} = \frac{4\text{m/s}}{5\,\text{m/s}} = 4/5$



29)

F.B.D of rope :

In horizontal direction

$T_A \cos 53 = T_B \cos 37$

$\Rightarrow T_A = \frac{4T_B}{3}$

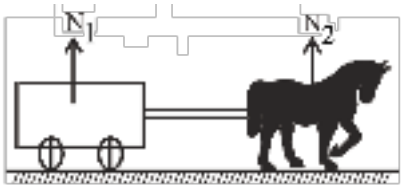
& in vertical direction

$T_A \sin 53 + T_B \sin 37 = 900$

$T_B = 540$ N

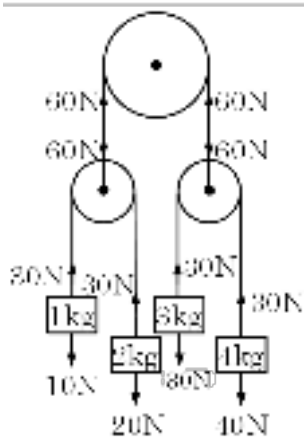
$T_A = 720$ N

Horizontal component of tension must be same at each point of rope but due to weight, vertical component of tension must be different.

30) 

Ground applies normal force on cart as well as on horse, which is an external force.

And force applied on cart by horse & on horse by cart are internal forcce. All statements are correct

31) 

On 1 kg; $30 - 10 = 1 \times a_1$

$\Rightarrow a_1 = 20$ m/s^2

On 2 kg; $30 - 20 = 2 \times a_2$

$\Rightarrow a_2 = 5$ m/s^2

On 3 kg; $30 - 30 = 3 \times a_3$

$\Rightarrow a_3 = 0$

On 4 kg; $40 - 30 = 4 \times a_4$

$\Rightarrow a_4 = 2.5$ m/s^2

32) $T_i = mg$

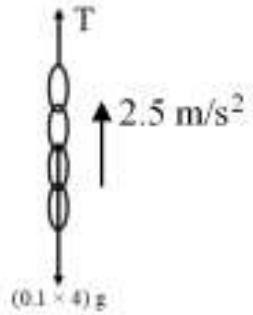
After placing extra mass m' :

$a = \frac{(m+m')g - mg}{2m+m'} = \frac{m'g}{2m+m'}$



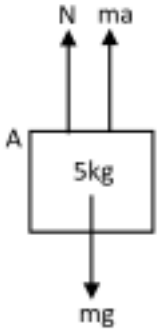
T' – mg = ma ⇒ T' = mg + ma = $\frac{4}{3}$mg

33) We are given a chain which is moving vertically upwards with constant acceleration 2.5 m/s^2 .The chain consists of 5 links of mass 0.1 kg each . We have to calculate the force of interaction between the top link and the link immediately below it. FBD of chain



T – (0.1 × 4)g = (0.1 × 4) × 2.5
T = 4.92 N

34) We need to determine the force excerted by block 'A' on block 'B' when the elevator is descending with an acceleration.
FBD of 5 kg block :

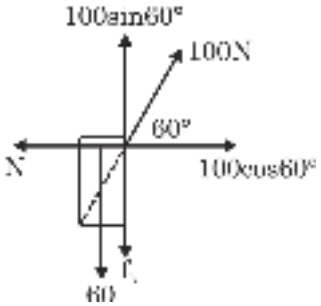


Since the elevator is accelerating downwards, the effective acceleration experienced by block A is (g–a). Therefore the Normal force N can be calculated as :
N = m (g–a) = 5 (10 – 2)
⇒ N = 40 newton.

35)

Limiting friction is unknown.

36) N = 100 cos 60° = 50 N
 $(f_s)_{max}$ = 0.5 × 50 = 25 N



$f_k = 0.3 \times 50 = 15$ N

Net external force, $F_{11} = 100 \times \frac{\sqrt{3}}{2} - 60$

$= 50\sqrt{3} - 60$

≈ 26.6 N > 25 N

 The block will move upward , so kinetic friction acts on it downward

37) $F_L = \mu_S$ mg

$200 = \mu_s \times 500$

$\mu_s = 0.4$

$f_k = \mu_k$ mg

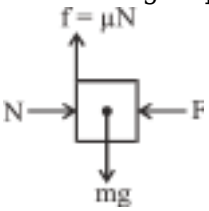
$150 = \mu_k \times 500$

$\mu_k = 0.3$

38) $a = \frac{16-4}{2} = 6 m/s^2$

39) for equilibrium N = F

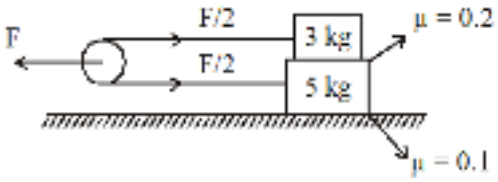
& f = mg or μN = mg



So μF = mg

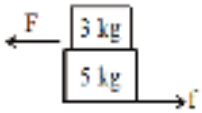
⇒ 0.25 F = 2 × 10

⇒ F = 80 N



40)

For common Acceleration



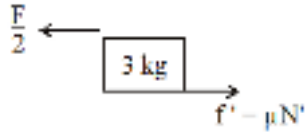
f = μN = 0.1 × 8g = 8N

F – f = Ma_C

$$a_C = \frac{F-f}{M}$$

$$a_C = \frac{F-8}{8} \quad .....(i)$$

For No slipping, Both Block should move with common acceleration



$f' = 0.2 \times 3g = 6N$

This is limiting force

$$\frac{F_{max}}{2} - f' = 3a_C$$

$\uparrow_{max}$

$$\frac{F_{max}}{2} - 6 = 3 \times \frac{F_{max}-8}{8}$$

$4F_{max} - 48 = 3$

$\boxed{F_{max} = 24}$

41) $u_y = u \sin a = 52 \times \frac{5}{13} = 20$ m/s

$s_y = 15, \quad \vec{a}_y = -g$

Now by using $s_y = u_y t + \frac{1}{2} a_y t^2$

$15 = 20t - 5t^2$

$5t^2 - 20t + 15 = 0$

$t^2 - 4t + 3 = 0$

$t = 1, 3$ therefore $\Delta t = 2$ sec.

42) The angle between gravitational force and velocity changes continously.

43) By F.B.D

$N_C > N_A = N_B$ is not correct

44) Initial velocity along X-axis and Y-axis are 3 and 4 ms^{-1} respectively. Acceleration along X and Y-axes = $\frac{2}{1}$ and $\frac{-3}{1}$ or 2 and –3 respectively.

$s_x = ut + \frac{1}{2} at^2 = 3 \times 3 + \frac{1}{2} \times 2 \times 9 = 18m$

$s_y = ut + \frac{1}{2} at^2 = 4 \times 3 + \frac{1}{2} \times -3 \times 9$

$= 12 - 13.5 = -1.5$ m

So required x and y-coordinate = (18, –1.5)

45) Let extension in the spring is x.

For 10 kg mass : $kx = 10a_1 = 10 \times 12 = 120$ N

For 20 kg mass : $200 - kx = 20a_2$

$\Rightarrow 200 - 120 = 20a_2 \Rightarrow a_2 = 4 m/s^2$



CHEMISTRY

46) **Question Explanation:** The question asks for the total number of atoms in 1 mole of $Ca(OH)_2$.
Concept: Number of atoms = mole × atomicity × Avogadro's number (6.02×10^{23})
Solution:
One formula unit of $Ca(OH)_2$ contains 1 calcium (Ca) atom, 2 oxygen (O) atoms, and 2 hydrogen (H) atoms, totaling 5 atoms per formula unit.
Number of atoms in 1 mole of $Ca(OH)_2$ = 1 mole × 5 atoms × Avogadro's number (6.02×10^{23}) = $5 \times 6.02 \times 10^{23}$ atoms.
Final Answer: Option (2)

47) **Explanation Question:** The question asks for the volume occupied by 4.8 grams of ozone (O_3) gas.
Concept: Number of moles and molar volume of a gas at STP .
Solution:
We fist find the number of moles by dividing the given mass by the molar mass of ozone (O_3). Then, we multiply the moles by the molar volume of a gas at STP (22.4 L).
Molar mass of O_3 = 16 x 3 = 48 g/mol
Moles = $\frac{4.8}{48} = 0.1$ mol
Volume = 0.1 x 22.4= 2.24 L
Final Answer: Option (2)

48) **Question Explanation :** Calculate the minimum molecular mass of an enzyme given Se's percentage.
Concept : Minimum mass requires at least one atom of Se in the molecule.
Solution :
% of an element = $\frac{\text{Atomicity} \times A_M}{\text{Minimum } M_M} \times 100$
0.5% = $\frac{1 \times 78.4}{\text{Minimum } M_M} \times 100$
Minimum molecular mass = $\frac{78.4}{0.5} \times 100$
$= 1.568 \times 10^4$ amu
Final Answer : Option (1)

49)

Total moles CO = $\frac{280 \times 10^{-3}}{28} = 10^{-2}$

Moles removed = $\frac{10^{21}}{6 \times 10^{23}} = \frac{1}{6} \times 10^{-2}$

Moles remain = $10^{-2} - \frac{1}{6} \times 10^{-2}$

$= \frac{5}{6} \times 10^{-2}$

50) $n = \frac{11.2}{22.4} = \frac{1}{2}$

$n = \frac{w}{M_w}$

$\frac{1}{2} = \frac{15}{M_w}$

$M_w = 30$

$M_w = 2 \times V.D.$

$30 = 2 \times V.D.$

$V.D. = 15$

51)

$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$

16 g $CH_4 \rightarrow$ 36 g H_2O

8 g $CH_4 \rightarrow$ 18 g H_2O

52)

Atomic weight of He = 4 amu

1 amu = 1.66×10^{-24} g

 mass of one He atom in gram

$= 4 \times 1.66 \times 10^{-24}$ g

$= 6.64 \times 10^{-24}$ g

53) **1. Explaining the question:**

This is an assertion-reason question about atomic mass units (amu).

2. Concept:

• 1 amu = 1/12th the mass of a carbon-12 atom = 1.66×10^{-27} kg

3. Solution:

Assertion is true

Reason is false 1 amu is not the actual mass of one atom

4. Final Answer:

Option 3. Assertion is true but Reason is false

54)

$n = m \times v$

$n = 0.24 \times \frac{500}{1000}$

$n = 0.12$

weight = 0.12 × 170 = 20.4 g

55)

$C_xH_y + \left(x + \frac{y}{4}\right) O_2 \rightarrow x\ CO_2 + \frac{y}{2} H_2O$

0.3 x = 2.4

x = 8

$0.3 \times \frac{y}{2} = 2.7$

y = 18

Hence hydrocarbon is C_8H_{18}.

56)

$$\% \text{ purity} = \frac{1.5}{2} \times 100$$
$$= 75$$

57) $\frac{x}{30} \Rightarrow y\frac{2x}{60} \Rightarrow \frac{x}{30} \Rightarrow y$

58) Moles of Zn $= \frac{9.81}{65.39} = 0.15$

Moles of Cr $= \frac{1.8 \times 10^{23}}{6.023 \times 10^{23}} = 0.3$

Moles of O = 0.6

So, whole molar ratio

= 0.15 : 0.3 : 0.6 = 1 : 2 : 4

Hence, formula = $ZnCr_2O_4$

59) **Explain Question:** Match amounts using the reaction: $C_3H_8(g) + 5O_2(g) \rightarrow 3CO_2(g) + 4H_2O(g)$.

Concept: This question is based on mass-volume relationship

Solution:

(a) 8 mol $H_2O \Rightarrow C_3H_8$ needed = 8/4 = 2 mol ⇒ volume = 2×22.4 = 44.8 L ⇒ (iii).

(b) 88 g C_3H_8 = 2 mol ⇒ O_2 needed = 2×5 = 10 mol ⇒ mass = 10×32 = 320 g ⇒ (iv).

(c) 44 g C_3H_8 = 1 mol ⇒ CO_2 formed = 3 mol ⇒ (ii).

(d) 1 mol C_3H_8 with 5 mol $O_2 \Rightarrow H_2O$ = 4 mol ⇒ mass = 4×18 = 72 g ⇒ (i).

(a)-(iii), (b)-(iv), (c)-(ii), (d)-(i)

Final Answer: 4

60) $\text{Molarity(M)} = \frac{\%\ \text{w/w} \times \text{density(g/ml)} \times 10}{\text{(Molar mass of solute)}}$

61) **Question Explanation**

The question asks how the equilibrium constant (K) of a reaction changes with an increase in temperature.

Concept

Van 't Hoff Equation / Le Chatelier's Principle (Effect of Temperature on Equilibrium)

Reasoning

The effect of temperature on the equilibrium constant depends on whether the reaction is endothermic or exothermic:

- For an endothermic reaction (ΔH > 0): An increase in temperature favors the forward reaction (product formation), so the equilibrium constant (K) increases.
- For an exothermic reaction (DH < 0): An increase in temperature favors the reverse reaction (reactant formation), so the equilibrium constant (K) decreases.

Therefore, the equilibrium constant may increase or decrease depending on the nature of the reaction (endothermic or exothermic).

Final Answer

Correct option: 4

62)

Equilibrium will shift in direction of more number of gaseous moles.

63)

Based on Le Chatelier's Principle (Effect of Pressure and Temperature on Equilibrium)

64) **Explain the question :** We need to find equilibrium constant (K_C) for given reaction.

Concept : This question is based on equilibrium constant (K_C)

Solution :

	A	+	B	⇌	C	+	D
t = 0	1		1		–		–
t_{eq}	1 – x		1 – x		x		x

At equilibrium : [C] = 2[A]

$x = 2(1 - x) = 2 - 2x$

$3x = 2$

$x = \frac{2}{3}$

$1 - x = 1 - \frac{2}{3} = \frac{1}{3}$

$$K_C = \frac{[C][D]}{[A][B]} = \frac{\frac{2}{3} \times \frac{2}{3}}{\frac{1}{3} \times \frac{1}{3}} = 4$$

$K_C = 4$

Final answer : 1

65)

Question Explanation : We need to compare K_p and K_c. Relation is

$K_p = K_c (RT)^{\Delta n}$

If $\Delta n < 0 \rightarrow K_p < K_c$.

Concept:- This question is based on Δn = moles of gaseous products - moles of gaseous

reactants.
Solution :-
1. $N_2O_3(g) \rightleftharpoons NO_2 + NO$
$\Delta n = (1 + 1) - 1 = +1 \rightarrow K_p > K_c$
2. $2H_2 + O_2 \rightleftharpoons 2H_2O$
$\Delta n = 2 - (2 + 1) = -1 \rightarrow K_p < K_c$
3. $H_2 + I_2 \rightleftharpoons 2HI$
$\Delta n = 2 - 2 = 0 \rightarrow K_p = K_c$
4. $2SO_3 \rightleftharpoons 2SO_2 + O_2$
$\Delta n = (2 + 1) - 2 = +1 \rightarrow K_p > K_c$
Final Answer : Option (2)
$2H_2(g) + O_2(g) \rightleftharpoons 2H_2O(g)$

66)

Explaining the Question :- We are comparing the stability of nitrogen oxides using their equilibrium constants of formation.
Concept:- This question is based on Larger equilibrium constant → product is more stable (formation favored).
Solution :- For NO: $K = 5 \times 10^{-31}$
• For NO_2 ; $K = 8 \times 10^{-31}$
• For N_2O_5 ; $K = 1 \times 10^{-34}$
Order of stability (higher K → more stable)
NO_2 (8×10^{-31}) > NO (5×10^{-31}) >> N_2O_5 (10^{-34})
Final Answer :- 4

67)

$$K_{eq} = \frac{K_f}{K_b} = \frac{0.25}{5000} = 5 \times 10^{-5}$$

68)

Explaining the Question :- We need to convert the given equilibrium reaction into the required one and find the new equilibrium constant.
Concept:- This question is based on
• If reaction is multiplied/divided, the equilibrium constant is raised to that power.
• If reaction is reversed, reciprocal of K is taken.
Solution :-
1. $2HCl \rightleftharpoons H_2 + Cl_2; K = 4 \times 10^{-34}$
2. Reverse reaction
$H_2 + Cl_2 \rightleftharpoons 2HCl$
$K' = 1/(4 \times 10^{-34}) = 2.5 \times 10^{33}$
3. Required reaction;
$$\frac{1}{2}H_2 + \frac{1}{2}Cl_2 \rightleftharpoons HCl$$
This is $\frac{1}{2}$ times the above reaction.
so, $K = (2.5 \times 10^{33})^{(1/2)}$

$= \sqrt{(2.5 \times 10^{33})}$
$\approx 5 \times 10^{16}$
Final Answer :- 4

69)

Question Explanation: Check correctness of Statement-I and Statement-II for equilibrium: $H_2O(s) \rightleftharpoons H_2O(\square)$
Concept: Le-Chatelier's principle
Solution:
• Melting of ice is **endothermic**.
• Ice has **greater volume** than liquid water.
• Increase in temperature favors **endothermic** direction.
• Increase in pressure favors the side with **smaller volume**.
Statement-I: Increasing temperature favors melting and increasing pressure favors liquid water (smaller volume) → **more water forms** → **correct**.
Statement-II: Forward reaction is endothermic (**true**), but it says volume of water is greater than ice (**false**, ice has greater volume). Hence overall **incorrect**.

Statement I is correct but Statement II is incorrect.
Final Answer: (Option 2)

70)

Question Explanation:
Interpret the meaning of a very small equilibrium constant (K_p = 0.0005 atm) for the reaction $A(g) \rightleftharpoons 2B(g)$.

Concept: Prediction of stability of reactant and product

Solution:

$K_p \square 1$ → equilibrium lies to the **left**, reactants predominate.

Products are **less stable**, reactants are **more stable**.

Small K_p → very little dissociation of reactants.

Relationship between K_p and K_c: $K_p = Kc(RT)^{\Delta n}$.

A. $\Delta n = 2 - 1 = 1 \rightarrow K_c = K_p / (RT) \rightarrow Kc < K_p$ at 300 K.

(I) Products more stable → incorrect (K_p small → reactants more stable)

(II) Reactants very little dissociated → Correct

(III) Reactants more stable than products → Correct

(IV) $K_c < K_p$ at 300 K → Correct

Final Answer: Option (3)

71)

Explain Question : We need Kp for $NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$, given total equilibrium pressure = 0.660 atm.
Concept : This question is based on equilibrium constant for heterogeneous equilibrium.
Solution : For $NH_4HS(s) \rightleftharpoons NH_3(g) + H_2S(g)$
Let equilibrium pressure of NH_3 = P and H_2S = P.
Total pressure = P + P = 0.660 → P = 0.33 atm.
$K_p = P_{NH_3} \times P_{H_2S}$
$K_p = (0.33)(0.33) = 0.1089 \approx 0.109$.
Final answer : option (1)

72) $K = \frac{(0.4)^2}{(0.1)} \Rightarrow 1.6$ M

73) $2AB(g) \rightleftharpoons 2A(g) + B_2(g)$

t = 0	1 mole		
t = eq	1–2x	2x	x

$$\alpha = \frac{1}{2} = \frac{2x}{1} \Rightarrow x = \frac{1}{4}$$

t = eq	$1-\frac{2}{4}$	$\frac{2}{4}$	$\frac{1}{4}$
	$=\frac{1}{2}$	$=\frac{1}{2}$	$=\frac{1}{4}$

$$(n_T)_{eq} = \frac{1}{2} + \frac{1}{2} + \frac{1}{4} = \frac{5}{4}$$

$$P_{B_2} = \frac{\left(\frac{1}{4}\right)}{\left(\frac{5}{4}\right)}.P = \frac{P}{5}$$

$$P_A = \frac{\left(\frac{1}{2}\right)}{\left(\frac{5}{4}\right)}.P = \frac{4}{10}P$$

$$P_{AB} = \frac{\left(\frac{1}{2}\right)}{\left(\frac{5}{4}\right)}.P = \frac{4}{10}P$$

$$K_p = \frac{P_A^2 \times P_{B_2}}{P_{AB^2}} = \frac{P}{5} \Rightarrow \frac{P}{K_p} = 5$$

74) **Explain the Question :** We need to find value of K_C for given reaction if value of K_P and T are given.
Concept: This question is based on relation between K_P and K_C.
Solution :
$CaCO_3 \rightleftharpoons CaO + CO_2$; $\Delta n_g = 1 - 0 = 1$
(s) (s) (g)
$K_P = K_C(RT)^{\Delta n_g}$
$K_C = \frac{K_P}{(RT)^{\Delta n_g}}$

$$K_C = \frac{167}{(0.08314 \times 1073)^1}$$

$$K_C = \frac{167}{88.20} = 1.896$$

$K_C = 1.896$ mol L^{-1}

Final Answer : (1)

75) At t = 3 min, according to graph
$[H_2]$ increases first ie, H_2 was added to the equilibrium

76)

Explain the Question : The problem gives number of electrons in X^{3+} and asks for protons in neutral atom X. Relation of charge, electrons and protons is used.
Concept : This question is based on Protons = Atomic number of the element.
Solution :
For cation : Electrons = Protons – Charge
Given, electrons in X^{3+} = 16
⇒ p – 3 = 16
⇒ p = 19
Final answer : 4

77) **Explain the Question :** Calculate the frequency of radiation for a wavelength of 1Å.
Concept : This question is based on wave relation - frequency and wavelength.
Solution : The relation between frequency (v), wavelength (λ) and speed of light (c) is :
$v = c/\lambda$
Where :
v = frequency of radiation (Hz)
c = speed of light = 3×10^8 m/s
λ = wavelength of radiation
Given :
l = 1Å = 1×10^{-10}m
Substitute the values :
$v = (3 \times 10^8)/(1 \times 10^{-10}) = 3 \times 10^{18}$ Hz
Since 1 Hz = $1s^{-1}$ = 1 cycle/sec, all representations are equivalent.
Final Answer : 4

78) **Explain Question :-** The question asks for the percentage abundance of the isotope C^{41} based on the given average atomic weight of element X.
Concept :-This question is based on Average Atomic weight
Solution :-
X^{40} X^{41} X^{42}
30% x 70–x

$$41.25 = \frac{40 \times 30 + 41x + (70 - x)42}{100}$$

4125 = 1200 + 41 x + 2940 – 42 x
x = 15
Final answer:- (3)

79)

Explain Question : Find nucleus diameter for mass number A = 125 using nuclear radius formula.
Concept : This question is based on nuclear radius relation $R = R_0A^{1/3}$
Explanation : $R = 1.33 \times 10^{-13} \times (125)^{1/3}$ cm $= 1.33 \times 10^{-13} \times 5$ cm $= 6.65 \times 10^{-13}$ cm
Diameter $= 2 \times R = 1.33 \times 10^{-12}$ cm
Final answer : 2

80) **Question Explanation :**
The question asks for the energy of an electron in the second orbit (n = 2) of a helium ion (He^+).

Concept :
The energy of an electron in the nth orbit of a hydrogen-like species is given by the formula $E_n = -13.6 \times \frac{Z^2}{n^2}$ eV.
Solution :
For He^+, the atomic number (Z) is 2. The electron is in the second orbit, so n = 2

Using the formula:

$$E_2 = -13.6 \times \frac{Z^2}{n^2}$$

$$E_2 = -13.6 \times \frac{2^2}{2^2}$$

$$E_2 = -13.6 \times \frac{4}{4}$$

$E_2 = -13.6$ eV

Final Answer : option (2)

81)

It explain the spectrum of atom or ion containing one electron.

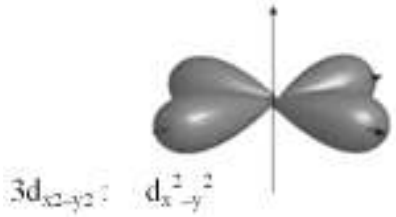
82) $T = \frac{2\pi r}{v} = \frac{2\pi \times n^2 h^2/4\pi^2 mze^2}{2\pi ze^2/nh}$ $T \propto n^3/z^2$

83) **Explain the Question :** A gas absorbs a photon of 355 nm and emits two photons; one emission is at 680 nm, find the wavelength of the other photon.
Concept : This question is based on energy conservation in photon emission.
Solution :
Energy of photon : $E = hc/\lambda$
Absorbed photon energy = Sum of emitted photon energies :
$hc / \lambda_0 = hc / \lambda_1 + hc / \lambda_2$
Divide both sides by hc :
$1/\lambda_0 = 1/\lambda_1 + 1/\lambda_1$
Substitute values :

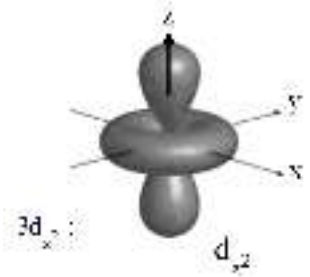
$1/\lambda_2 = 1/\lambda_0 - 1/\lambda_1$
$1/\lambda_2 = 1/355 - 1/680$
$1/\lambda_2 = 0.0028169 - 0.0014706 = 0.0013463$
Take reciprocal :
$\lambda_2 = 1/0.0013463 \approx 743$ nm
Final answer : 3

84)

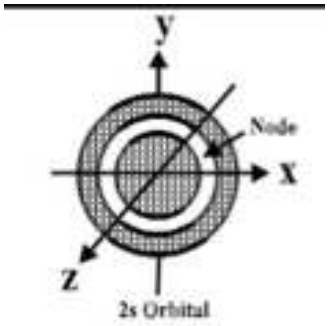
Explain Question : Identify which one of the four statements about orbital nodes/electron density is correct.
Concept : This question is based on nodes in atomic orbitals.
Solution :



electron density in xy plane is non-zero (lobes lie in xy)

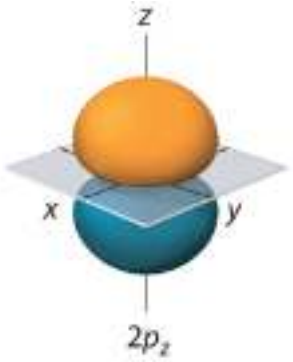


electron density in xy plane is nonzero , due to presence of electron ring in xy plane.



2s :

has one spherical (radial) node



$2p_z$:
nodal plane is xy , not yz
Final answer : option (3)

85) **Explain Question:**
We have to identify the incorrect statement related to Bohr's model energy expressions for hydrogen-like species.
Concept:
For a hydrogen-like ion:

- Total energy: $E_n = \frac{-2\pi^2 mk^2 z^2 e^4}{n^2 h^2}$
- Kinetic energy: $KE = -E_n$
- Potential energy: $PE = 2E_n$ (negative)

Solution:
Statement 1:
For Li^{2+} (Z = 3), n = 3

$$E_3 = \frac{-2\pi^2 mk^2 e^4 \times 9}{9h^2} = \frac{-2\pi^2 mk^2 e^4}{h^2}$$ Correct.

Statement 2:
For H-atom, n = 2

$$E_2 = -\frac{13.6}{4} = -3.4eV$$

$PE = 2E_2 = -6.8$ eV
Given statement says 6.8 eV (sign ignored) Incorrect.
Statement 3:
For He^+ (Z = 2), n = 3

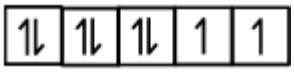
$$KE = -E_3 = \frac{2\pi^2 mk^2 e^4 \times 4}{9h^2} = \frac{8\pi^2 mk^2 e^4}{9h^2}$$ Correct.

Final Answer: Option(2)

86)

$X^+ = 22\ e^-$, 23p
$Y^{2+} = 22\ e^-$, 24p
$Z^- = 22\ e^-$, 21p

87) (a) $Ni^{+2}[Ar]4s^2 3d^8$



(b) Cr = $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1\ 3d^5$
l = means → subshell ⇒ $7e^-$
(c) Mg = $1s^2\ 2s^2\ 2p^6\ 3s^2$ penultimate shell contains $8e^-$
(d) e^-s not present in the nucleus

88)

Question Explanation: Using Heisenberg uncertainty principle with de-Broglie relation.

Concept:

A. de-Broglie wavelength: $\lambda = h / (mv)$
B. Uncertainty principle: $\Delta x \cdot \Delta p \geq h / (4\pi)$
C. Given: $\Delta x = \lambda / (4\pi)$

Solution:
$\Delta x \cdot \Delta p = (\lambda / 4\pi) \cdot \Delta p = h / 4\pi \Rightarrow \Delta p = h / \lambda$

But $p = mv = h / \lambda \Rightarrow \Delta p = p$

Thus, $\Delta v = \Delta p / m = p / m = (mv) / m = v$
Final Answer: Option (2)

89) $\bar{v} = \frac{1}{\lambda} = RZ^2 \left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$ ($n_1 = 2$, $n_2 = n$)

$$\bar{v} = R \times (1)^2 \left[\frac{1}{(2)^2} - \frac{1}{n^2}\right]$$

$$\bar{v} = R\left[\frac{1}{4} - \frac{1}{n^2}\right] \Rightarrow \bar{v} = R \times \left[\frac{n^2 - 4}{4n^2}\right]$$

$$\boxed{\bar{v} = R\frac{(n+2)(n-2)}{4n^2}}$$

90)

$$\Delta x.\Delta p = \frac{h}{4\pi}$$

$$\Delta x.m\Delta v = \frac{h}{4\pi}$$

$$\Delta v = 5 \times 10^6 \times \frac{0.02}{100}$$

$$\Delta v = 1000$$

$$\Delta x = \frac{h}{4\pi.m\Delta v}$$

$$\Delta x = \frac{6.63 \times 10^{-34}}{4 \times 3.14 \times 9.1 \times 10^{-31} \times 1000}$$

$$\Delta x = 58 \times 10^{-9}\ m$$

91) **Explain Question:** Question asks to identify a given figure and select correct statements about it.
Concept: This question is based on Euglenoids
Solution: The figure shown is Euglena. Euglena has two flagella (one long, one short).
Final answer: 3

92)

Explain Question: Identify the given diagram and its class.

Concept: This question is based on Basidiomycetes

Solution: The diagram = *Agaricus*, and its class = Basidiomycetes.

Final answer: 2

93)

NCERT, Pg. # 17, 18

94) **Explain Question :** The question is asking us to identify correctly labelled parts (A, B, C, D) of a bacteriophage.
Concept : This question is based on Virus.
Solution : A = Head, B = Sheath, C = Collar, D = Tail fibres
Final answer : option (3)

95)

Explain Question: Question asks to identify the organism shown in the diagram and choose the correct statement about it.

Concept: Cynobacteria

Solution: The diagram shows a filamentous cyanobacterium with heterocysts, typical of Nostoc. It is a nitrogen-fixing cyanobacteria (blue-green algae).

Final answer: 1

96) NCERT-XI, Pg. # 91

97) **Explain Question -** The question asks which of the given taxa is not related to wheat.
Concept - Taxonomic Categories & Classification of Organisms
Solution -

Wheat belongs to Kingdom Plantae, Class Monocotyledonae, Family Poaceae, Genus Triticum.
Sapindales is an order that includes plants like neem, mango, etc., but not wheat.
Final Answer - Option 1

98)

Explain Question: The question asks about the relation which is similar to the relation of Solanaceae and Convolvulaceae with Polymoniales.

Concept: Order/ Family.

Solution:

The relation of Solanaceae and Convolvulaceae with Polymoniales is similar to the relation occuring in : Felidae and Canidae with Carnivora

So, the correct answer is: 1.

Final answer: Option (1)

99) NCERT Pg. # 8

100) **Question Explanation**: Question is asking about which algae forms motile colony
Concept: Algae-General Characters.
Solution:1.Volvox – Correct
Volvox is a green alga.
It forms a motile colony — a spherical colony of thousands of cells.
The colony can move using flagella present on individual cells that coordinate together.
This is a classic example of a motile colonial alga.
Final answer: 1

101)

Explain Question: Question asking about classification of algae
Concept: This question is based on Algae -classification.
Solution: The algae are divided into three main classes: Chlorophyceae, Phaeophyceae and Rhodophyceae on the basis of pigments and stored food
Final Answer: (4)

102)

Explain Question : Question asking about Asexual reproduction in liverworts by the formation of specialised structures called **gemmae**

Concept : This question is based on liverwort

Solution :
Asexual reproduction in liverworts takes place by the formation of specialised structures called **gemmae** (sing. gemma). Gemmae are green, multicellular, asexual buds, which develop

in small receptacles called gemma cups located on the thalli. The gemmae become detached from the parent body and germinate to form new individuals

Final Answer : option (2)

103) NCERT-XI, Pg. # 33, Para-02

104)

NCERT XI, Page No. # 32

105) **Solution :** The following characteristics are found in rhodophyaceae.
Final Answer : (3)

106)

Question Explanation: The question is asking about a statement based on the given aquatic fern (pteridophyte).



Concept: Pteridophyte Characteristics : *Salvinia*
Solution:
1. Considered as precurssor to seed habit
2. It is a homosporous plant – Incorrect. Aquatic ferns (like Salvinia) are heterosporous (produce two types of spores).
3. Sporangia formed in Ascocarp – Incorrect. Ascocarp is a fungal structure, not present in pteridophytes.
4. Belong to the smallest group of pteridophytes – Incorrect. Salvinia belongs to class Pteropsida . This is the largest group of pteridophytes, not small.
Final Answer: Option (1)

107)

Explain Question : The question asks you to identify how many plants from the provided list are non-vascular.

Concept : This question is based on non-vascular (Algae and bryophytes).

Solution :
The non-vascular plants in the list are Spirogyra, Volvox, Polysiphonia, Laminaria, Sphagnum, Marchantia, Funaria, Riccia, Moss, and Polytrichum, totaling ten plants.

Final Answer : option (1)

108)

NCERT XI Page No # 32,33,34
Green Algae
Spirogyra, Ulothrix, Volvox, Chara
Brown algae
Sargassam, Laminaria, Ectocarpus, Fucus, Dictyota

109) **Explain Question:** The question asks the link between Prokaryotes and multicellular eukaryotes.
Concept: Protista
Solution: Protista forms the Link between procaryotes and multicellular eucaryotes. So, the correct answer is: 2.
Final answer: 2.

110) NCERT-XI, Pg. # 27

111) **Question Explanation:** The question is asking about correct statement.
Concept: Kingdom Monera (Eubacteria), History, Shape, Cell Structure, Flagella, Pili.
Solution: Correct statements:
(A) Kingdom Monera includes only bacteria.
(B) Bacterial are found everywhere.
(C) Bacteria are smaller than eukaryotic cells.
(D) Mesosomes are helful in respiration and cell division.
(E) Bacteria show the most extensive metabolic diversity and can be heterotrophic, chemolithotrophic, Photoautotrophic, etc.
Answer (4)

112) NCERT Pg. # 11

113) **Question Explanation :** This question is asking about match the column.
Concept : History of taxonomy.
Solution :

Column-I		**Column-II**	
A	Binomial nomenclature	(i)	Linnaeus
B	Father of biology & zoology	(ii)	Aristotle
C	Darwin of 20th century	(iii)	Ernst Mayr
D	Father of botany	(iv)	Theophrastus

Final Answer : Option (1)

114) NCERT-XI, Pg # 94, 95

115)

NCERT-XI, Pg # 100, 101

116)

NCERT-XI, Pg # 96, 98

117) NCERT-XI, Pg # 95

118) NCERT-XI, Pg # 100, 101

119)

NCERT XI, Pg. # 99

120) NCERT XI, Pg. # 100, 101, 126

121)

NCERT XI, Pg. # 98, 87, 88

122)

NCERT XI, Pg. # 97, 98

123)

NCERT XI, Pg. # 95, 96, 97, 98

124) NCERT XI, Pg. # 96

125) NCERT XI, Page # 88

126)

NCERT XI, Page # 101

127)

NCERT XI, Page # 93

128)

NCERT XI, Page # 127, 128

129)

NCERT XI, Page # 127

130)

NCERT XI, Page # 126

131)

NCERT XI, Page # 124

132)

NCERT XI, Page # 126

133) NCERT XI, Page # 127

134)

NCERT XI, Page # 121, 122

135) NCERT-XI, Pg. # 126, 127, 128

136) XI NCERT, Pg. # 101, 102

137) XI NCERT Pg. No :- 102

138) NCERT XI Page No. 101

139) NCERT Pg. No. 104

140) NCERT, Pg. # 81

141) **Question Explanation:** We must find the correct tissue–function/organ match.
Concept: Frog - digestive and respiratory system

Solution:
(1) A-Gall bladder → Production of bile (Incorrect; gall bladder stores bile, does not produce it)
(2) B-Liver→ Incorrect (they are lungs not liver)
(3) D-Kidneys → Ducts from it opens in rectum (Incorrect; frog's ureters open into cloaca, not directly into rectum)
(4) D-Kidneys → Possess uriniferous tubules (Correct; kidneys of frog have uriniferous tubules for urine formation)
Final answer: D-Kidneys - posses uriniferous tubules

142)

Explain Question : This question is asking about identifying the type of connective tissue shown in diagram (A) and choosing the correct related statement.
Concept : Structural Organisation in Animals – Connective Tissues (Dense Regular & Dense Irregular C.T.)
Explanation : Diagram (A) shows dense regular connective tissue in which collagen fibres are arranged parallel, typical of tendons. Hence, the correct statement is that A is an example of a tendon.
Final answer :
Option 3

143)

NCERT XI, Pg. # 102

144) NCERT Pg # 103

145) NCERT Pg. # 82

146) Module

147) NCERT, Pg. # 82

148) NCERT Pg. # 82

149) Module

150) NCERT Pg. # 83

151) NCERT Pg. # 81

152) Module
Main function of compound epithelium is to provide protection against chemical and mechanical stresses.

153) NCERT, Pg. # 103

154) NCERT Pg. # 101

155) Module

156) Para–1, Page No. # 54

157) NCERT Page No. 42

158) NCERT Page No. 42

159) NCERT-XI, Pg. # 48,49

160) NCERT-XI, Pg. # 40,44,45

161) NCERT-XI, Pg. # 37

162) NCERT-XI, Pg. # 49

163)

NCERT-XI, Pg. # 56

164) NCERT Pg # 56

165) NCERT-XI, Pg. # 51

166) NCERT-XI, Pg. # 41

167) NCERT-XI, Pg. # 48

168) NCERT-XI, Pg. # 40

169) NCERT-XI, Pg. # 38

170) NCERT-XI, Pg. # 49 & 50

171) NCERT-XI, Pg. # 46

172) NCERT-XI, Pg. # 44

173)

NCERT-XI, Pg. # 44

174)

NCERT-XI, Pg. # 48 & 49

175)

NCERT-XI, Pg. # 48

176)

NCERT-XI, Pg. # 43

177) In the skull of fishes, reptiles and Aves has only one occipital condyle is present whereas in the skull of Amphibia and mammalia have has two occipital condyles.

178) NCERT-XI, Pg. # 43

179) NCERT-XI, Pg. # 38 & 39

180) NCERT-XI, Pg. # 40